

Edge Computing Accelerators Outline

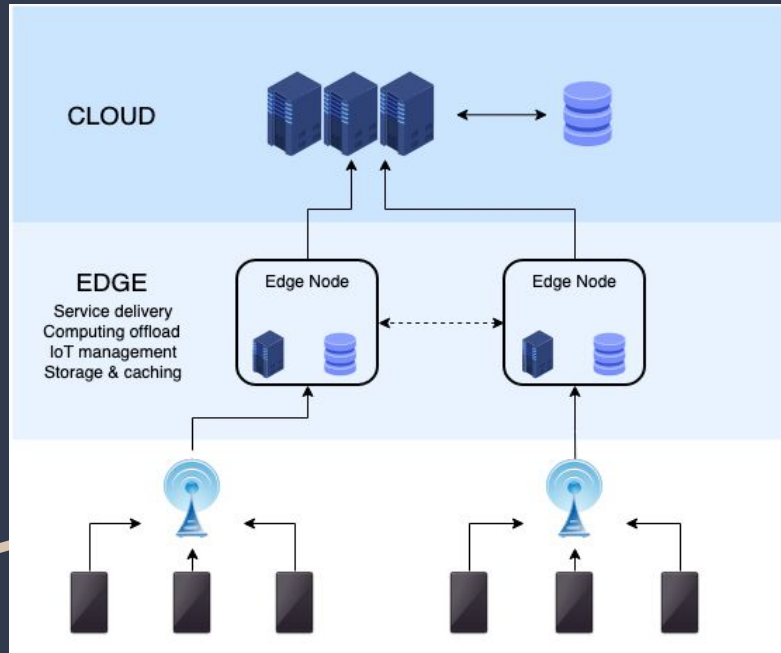
1. Introduction to Edge Computing in IoT (Sam)
2. What Are Edge Computing Accelerators? (Elijah)
3. Common Architectures & Technologies (Tanner)
4. Applications and Real-World Use Cases (Jesse)
5. Challenges, Trade-offs, and Future Trends (Lucas)

Edge Computing Accelerators for IoT Devices

Tanner, Jesse, Lucas, Elijah, Sam

A dark blue diagonal gradient bar that starts from the bottom left and extends towards the top right, covering the lower half of the slide.

Introduction to Edge Computing in IoT



What are IoT devices?

IoT devices are interconnected physical objects embedded with sensors, software, and connectivity that allow them to collect and exchange data.

What is Edge Computing?

Computing model that processes data near the source (IoT or Local Servers). Enabling fast, real-time decision making.

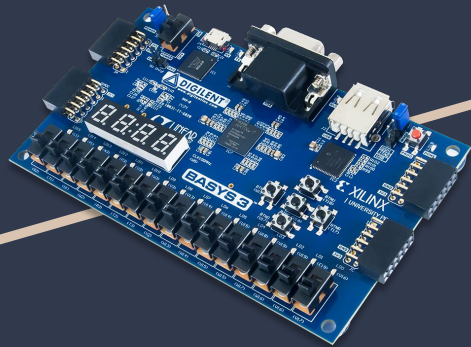
How does this work?

1. Sensors collect raw data (temperature, heart rate, vibration, etc.)
2. A nearby edge device (embedded processor, gateway, PU's, or microcontroller) processes that data
3. The system makes immediate decisions locally
4. Only important or summarized data gets sent to the cloud

Why IoT devices benefit from local processing?

- Reduced Bandwidth Usage
- Improved Data Security
- Better Scalability

What Are Edge Computing Accelerators?



Edge Computing Accelerators are specialized components designed to handle computationally-intensive tasks like matrix operations or inference with greater speed and energy efficiency compared to general-purpose CPUs.

Common types include:

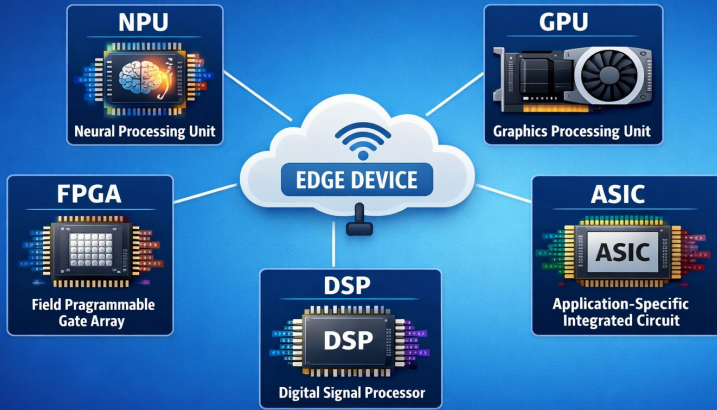
- Graphics Processing Units (GPUs)
- Field-Programmable Gate Arrays (FPGAs)
- Application-Specific Integrated Circuits (ASICs)

Why use Computing Accelerators?

- Enables efficiency and performance while working at the edge.
- Edge Devices, like smartphones, drones, or IoT sensors, can run complex models locally without relying on cloud servers.
- Reduces Latency and Improves Privacy.
- Allows AI applications to function with limited connectivity.

Common Architectures & Technologies

Edge Computing Accelerators



AI Generated Image

(Flexible) GPU – DSP – FPGA – NPU – ASIC (Efficient)

Graphics Processing Units (GPUs): Processors designed for parallel computation, primarily graphics and data-intensive processing.

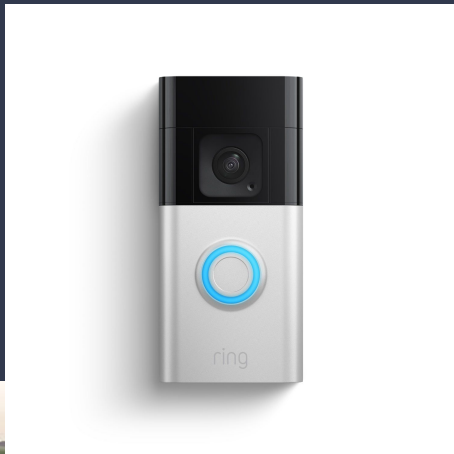
Digital Signal Processors (DSPs): Microprocessors optimized for real-time audio, video, and signal processing.

Field-Programmable Gate Arrays (FPGAs): Reconfigurable devices that implement custom hardware for parallel, low-latency processing.

Neural Processing Units (NPUs): Processors designed for efficient use in neural networks, artificial intelligence, and machine learning.

Application-Specific Integrated Circuits (ASICs): Custom-designed chips optimized for a specific task, improving efficiency and performance on the intended use case.

Applications and Real-World Use Cases



Security camera motion detection (image processing)

- Using a GPU/VPU for local processing before leaning on the network

Autonomous driving/navigation (neural models)

- Decision making through deep models run locally through GPUs and TPU (Tensor Processing Unit)

Center pivot irrigation

- Low speed/inconsistent network requires localized computing



The Future of Edge Computing



- AI pushing compute to the edge
- Security as a constant: focus shifts
- Latency down, complexity up
- Bandwidth reduced... then reintroduced?
- Hardware and Infrastructure challenges
- Hybrid solutions

HYBRID CLOUD

ENGINEERING GODS

Quiz Time!

Which of the following is NOT a benefit of edge computing for IoT devices?

- A. Reduced bandwidth usage
- B. Improved data security
- C. Lower power consumption
- D. Increased reliance on continuous internet connectivity



T/ F - Running AI models at the edge always produces more accurate results than running them in the cloud.

False (Edge models are often smaller and less accurate)

Quiz Time!

Which tradeoff is most directly associated with deploying AI inference at the edge rather than in the cloud?

- A. Increased latency but improved scalability
- B. Reduced latency but increased system complexity 
- C. Reduced compute requirements but increased bandwidth usage
- D. Increased security but reduced flexibility

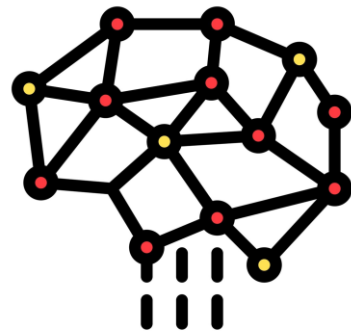
Which of the following is an example of an Edge Computing Accelerator?

- A. The North Bridge
- B. Graphics Processing Units (GPUs) 
- C. L1 Cache
- D. Central Processing Units (CPUs)

Neural Network ASICs

Team 2

Cameron Samson, Cecilee Blackburn,
Matthew Spotten, Michael Zaugg, Caeson
Reckling

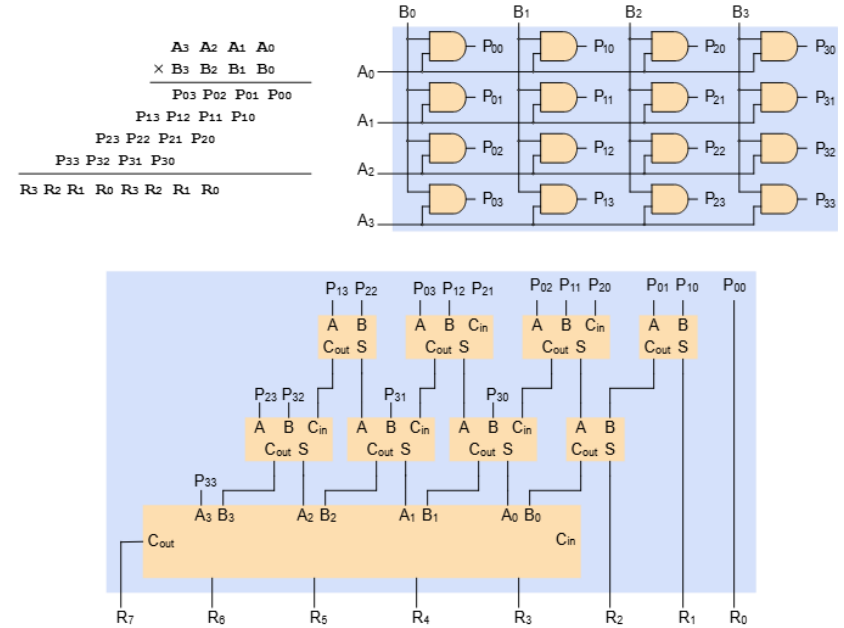


Topics - Neural Network ASIC

- Overview and Characteristics of ASICs
- Overview of Architecture
- Data Flow Architecture & Memory Hierarchy
- Not a need for high precision/Parallelism Strategies: data, model, pipeline
- Different types of neural networking ASICs

Overview of ASICs

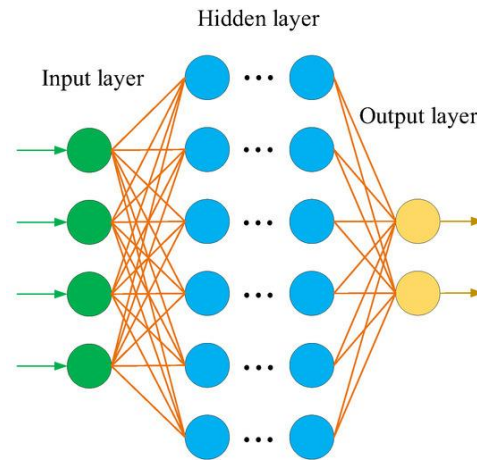
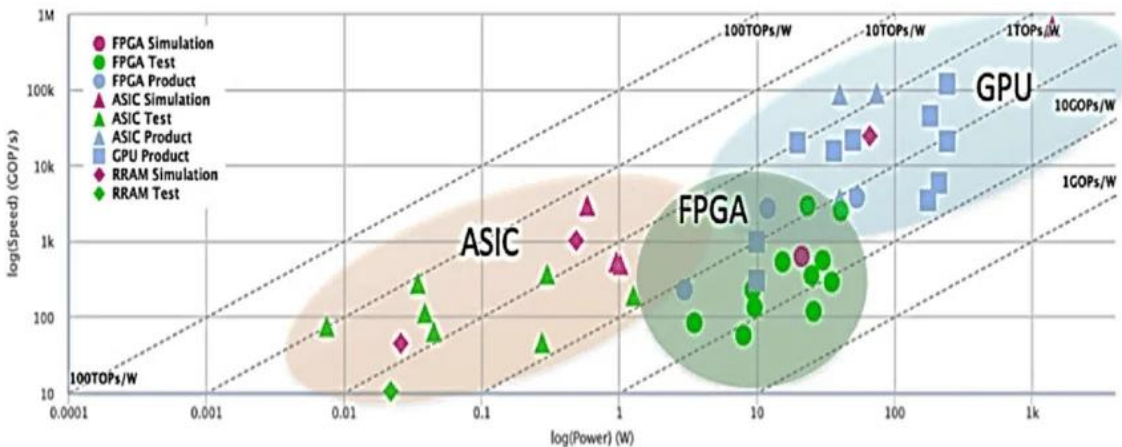
- Application-Specific Integrated Circuit.
- Designed to do exactly one task with great efficiency - water kettle vs stovetop.
- High upfront cost for design.
- High power efficiency.



Using ASICs for Neural Networks

ASICs are being used in the training and deployment (or inference) of Neural Networks.

ASICs lack the flexibility of GPUs and FPGAs as they cannot adapt to the changes of newer neural network architectures.



ASIC Architecture

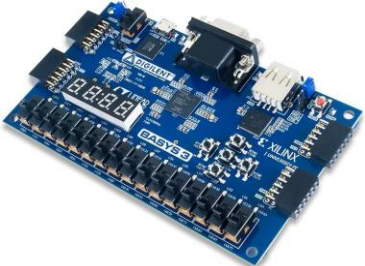
Full-custom design: Completely custom built ASIC using only the required components, offering maximum performance and smallest size but with high cost and long development time.

Semi-custom ASIC: Uses pre-designed transistor layers and customizes only the metal connections, reducing cost and speeding up development while maintaining some flexibility.

Standard cell-based ASIC: Constructed from predesigned logic blocks like gates and flip flops, balancing performance with faster design time and lower risk.

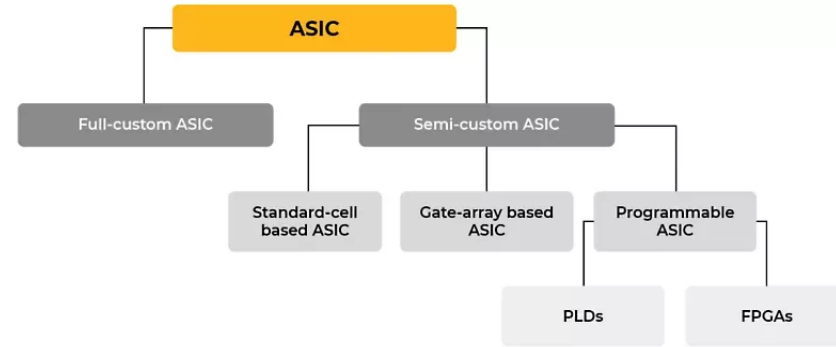
Gate-array based ASIC: Uses a pre-made array of transistors where only the wiring is customized, allowing quicker production with moderate flexibility.

Programmable ASIC: Can be configured after manufacturing (such as FPGAs), providing high flexibility but generally lower performance than fixed ASICs.



From Google

Types of ASIC Design



From Google

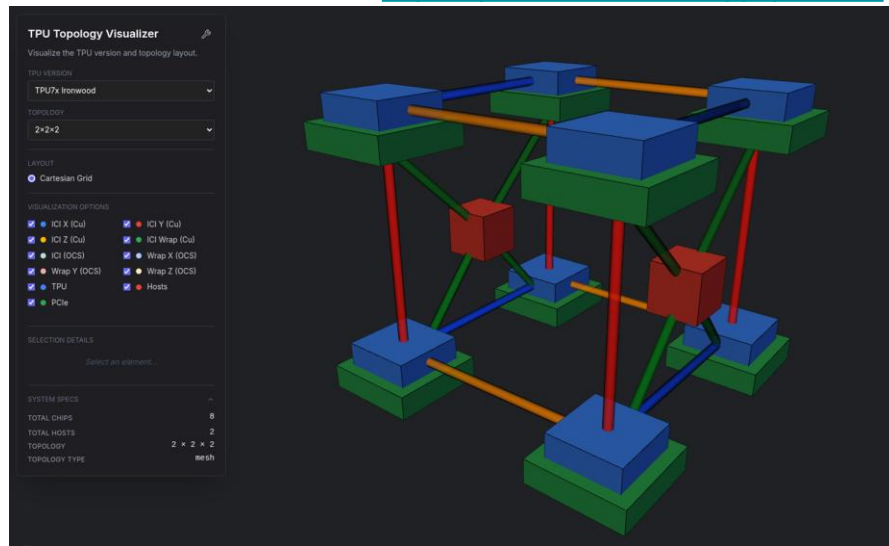
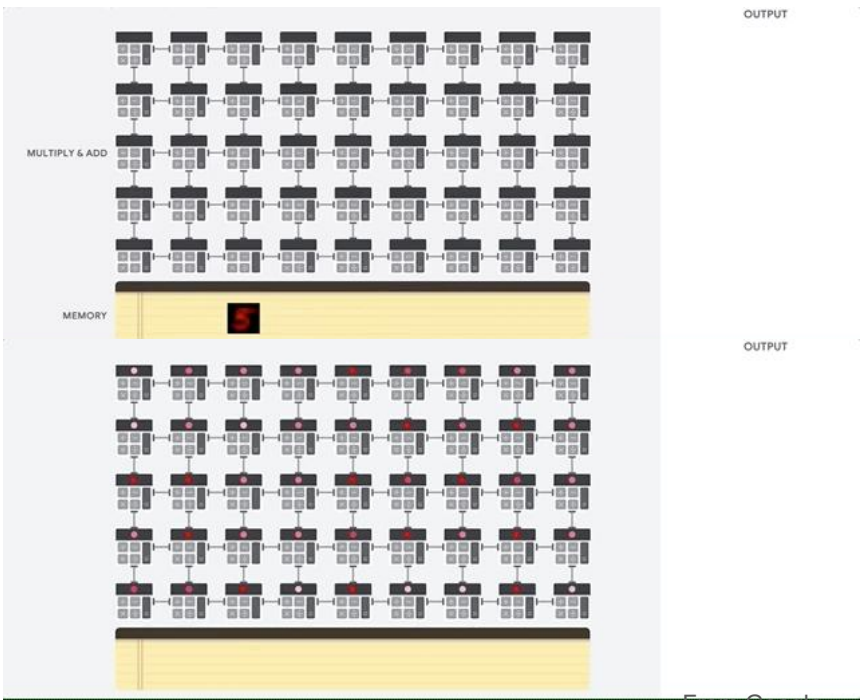
ASIC vs FPGA

Attribute	ASIC	FPGA
Functionality	Fixed after fabrication	Reprogrammable after manufacturing
Performance	Optimized for specific tasks	General-purpose, lower performance
Power Efficiency	Lower power usage	Higher power consumption
Use Case	High-volume, application-specific	Prototyping, flexible logic
Cost	High upfront cost, low per-unit cost at volume	Low upfront cost, high per-unit cost at volume

Architecture of ASIC - Google TPU Example

The primary task for TPUs (Tensor Processing Units) is matrix processing, which is a combination of multiply and accumulate operations. TPUs contain thousands of multiply accumulators that are directly connected to each other to form a large physical matrix.

<https://tpu-visualizer.uc.r.appspot.com/>

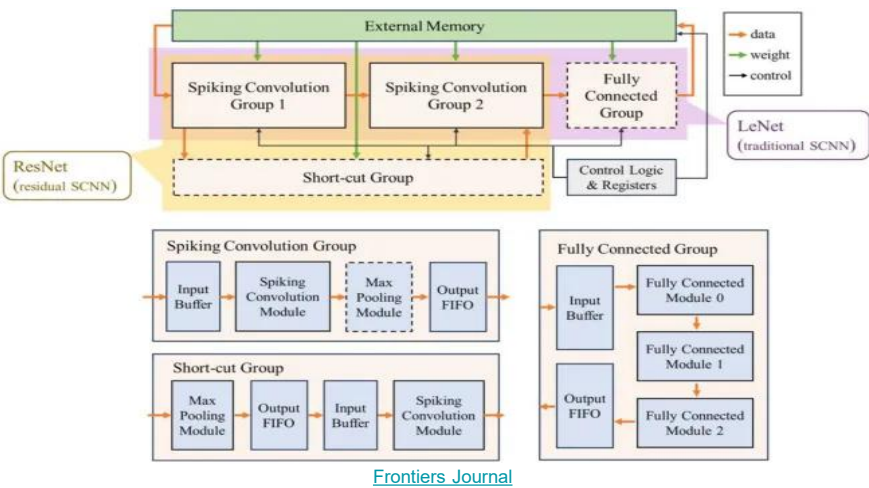


The TPU loads the parameters from HBM (High Bandwidth Mem) into the Matrix Multiplication Unit (MXU)

As each multiplication is executed the result is pipelined with the output being the summation of all results between the data and parameters.

Dataflow Architecture

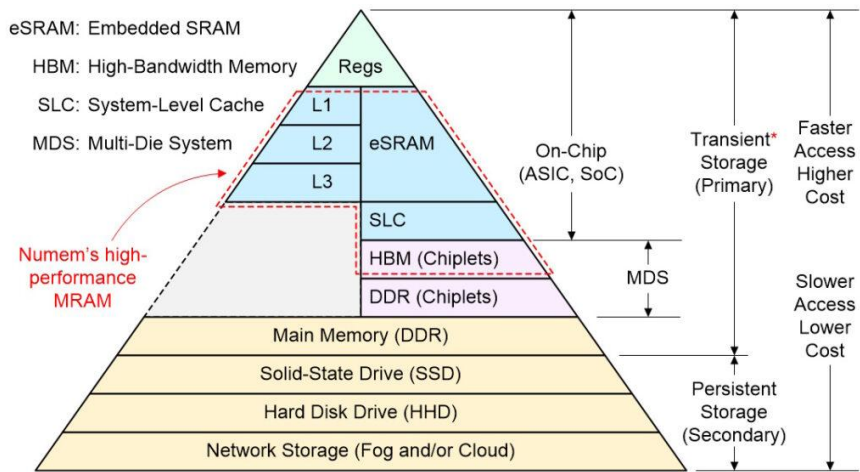
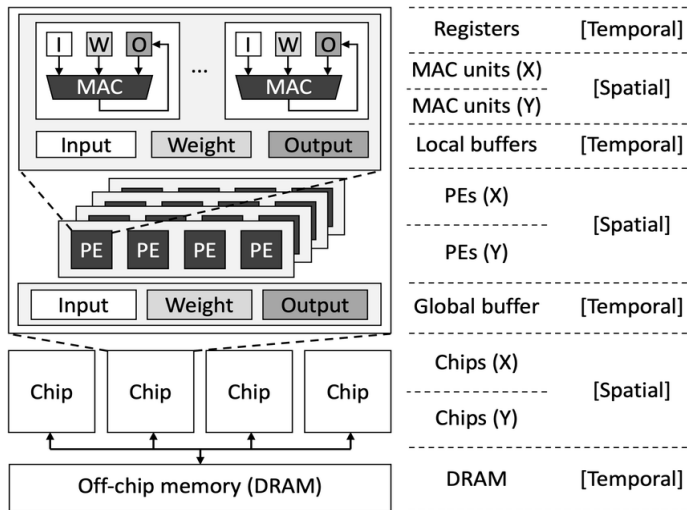
- Dataflow Architecture is the secret behind why modern NN ASICs can outperform general-purpose CPUs or GPUs.
- Traditional architectures focus on instruction-level-parallelism.
- Dataflow architecture focuses on data level parallelism.
- The operation executes as soon as its required operands are available, it doesn't wait for the program counter to reach a specific line in code.



Strategy	Hardware Setup	Data Movement Pattern	Efficiency Use
Weight Stationary	Weights are "locked" in PE Registers.	Activations "slide" horizontally; Partial Sums "drop" vertically.	TPU v1: Best for large batches.
Output Stationary	Partial Sums "stay home" in the Accumulator.	Weights and Activations stream from opposite sides.	Best for Bandwidth: Minimizes writing to DRAM.
Row Stationary	PEs handle 1D segments of a 2D convolution.	Highly complex 2D reuse of weights, pixels, and sums.	Eyeriss: The most energy-efficient.

Memory Hierarchy

The Component	Where it is in the Image
MAC Unit	The dark trapezoid labeled "MAC" .
Local Scratchpad	The boxes labeled I (Input), W (Weight), and O (Output).
NoC Router	The lines connecting the PE boxes .
Control Logic	The "dotted" hierarchy lines.



EE Journal: Max Maxfield

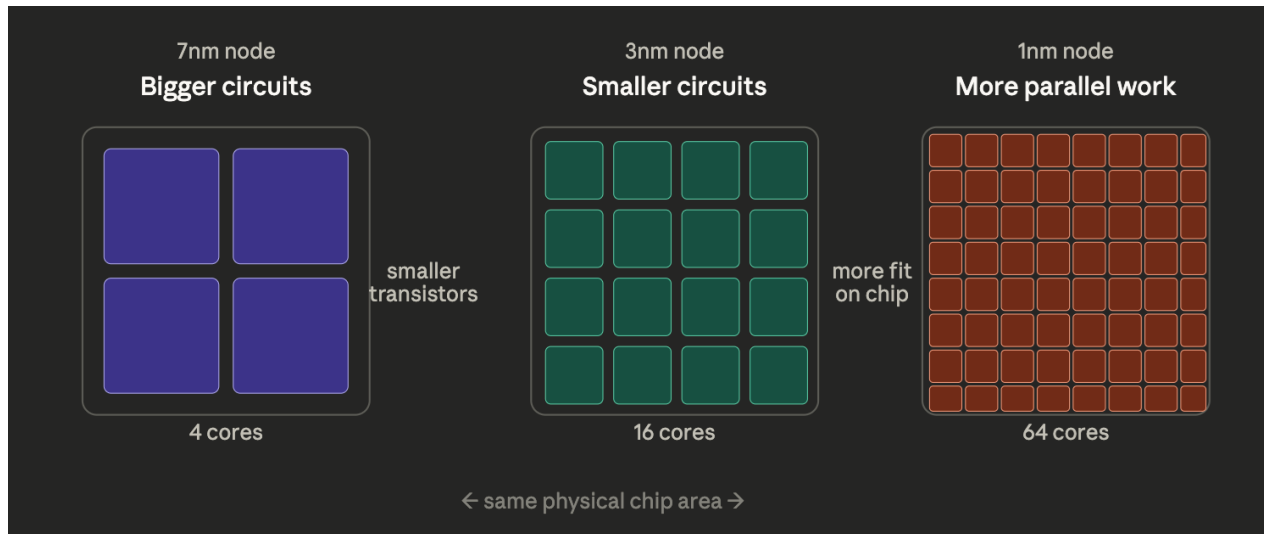
[NeuroSpector: Systematic Optimization of Dataflow Scheduling in DNN Accelerators](#)

- **Memory Gap & MRAM:** SRAM/DRAM bottleneck AI; MRAM provides a dense middle layer to reduce dataflow stalls.
- **Registers (L0):** Inside PEs; hold operands for MACs with minimal energy cost.
- **HBM (Middle Tier):** High-bandwidth memory that feeds PEs with weights/activations at high throughput.
- **Multi-Die Systems (MDS):** Chiplet-based design to scale computation + memory efficiently with better yield.
- **Dataflow vs Kernel Execution:** Dataflow keeps data local and moving through PEs; avoids costly DRAM round-trips.

No Need For High Precision

- Normal Computers
 - Need precise numbers
- AI chips
 - Use simpler numbers
- Smaller numbers → smaller circuits

AI generated



Parallelism in ASICs

- Data Parallelism - Cookie example
 - Simple to implement
 - Scales easily
 - Most common on ASICs
- Model Parallelism - Car example
 - Requires a lot of communication between chips
 - Needed when the model is too large to fit on one chip's memory
- Pipeline Parallelism
 - When you want HIGH throughput
 - Often combined with other methods



[dreamstime.com](https://www.dreamstime.com/illustration/Teamwork-people-working-together-assembly-puzzle-image-image41234567.html)

Types by Function

- Training:
 - HPC for training models, requires high precision and very large computation capacity
- Inference:
 - Pretrained models taking in new data
 - Reduced precision improves data movement speed
- Natural Language Processing:
 - Speech recognition, sequence based data, GPT models
- Vision Processing:
 - Specialized for convolutional neural networks, hardware for pixel processing, motion tracking, etc.

Architectural Approaches

- **Systolic Arrays:**
 - Lots of little Matrix Multiplication Units, data flows like a “heartbeat”
- **Neuromorphic Chips:**
 - Brain inspired circuits that mimic the behavior of neurons and synapses
 - Used in autonomous vehicles for environment adaptation
- **Analog ASICs:**
 - Perform calculation on analog signals and convert to digital data values
 - Much higher energy efficiency
- **Wafer Scale Engines:**
 - An entire wafer is used for one chip, reduces communication between others smaller chips

Questions

Question: What best describes data parallelism?

- A. Splitting a model into different parts and running each part on a different processor
- B. Running different models on the same data
- C. Running the same model on different pieces of data at the same time
- D. Processing one input at a time through multiple stages

True/False: TPUs are used mainly for matrix processing.

What is the difference between ASIC and FPGA?

- A. ASICs are reprogrammable after manufacturing, FPGAs are fixed.
- B. ASICs are optimized for specific tasks with lower power usage. FPGAs are flexible and re-programmable with less efficiency.
- C. ASICs have a lower upfront cost and higher per-unit cost, FPGAs have higher upfront cost and lower per-unit cost.
- D. ASICs are mainly used for prototyping, FPGAs are used for high volume production.

True/False: Dataflow architecture focuses on data-level parallelism rather than instruction-level parallelism. (Extra Question)

In dataflow systems, execution begins when:

- A. Clock cycle starts
- B. Instruction is decoded
- C. Operands are available (Answer)
- D. Memory is full

Questions - Answers

Question: What best describes data parallelism?

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T/F: Dataflow architecture focuses on data-level parallelism rather than instruction-level parallelism. **(True) (extra question)**

MCQ: Main Question for dataflow architecture

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- B. Instruction is decoded
- C. Operands are available (Answer)**
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Sources

[ASICs](#)

[Architecture](#)

[More Architecture](#)

[Data Parallelism](#)

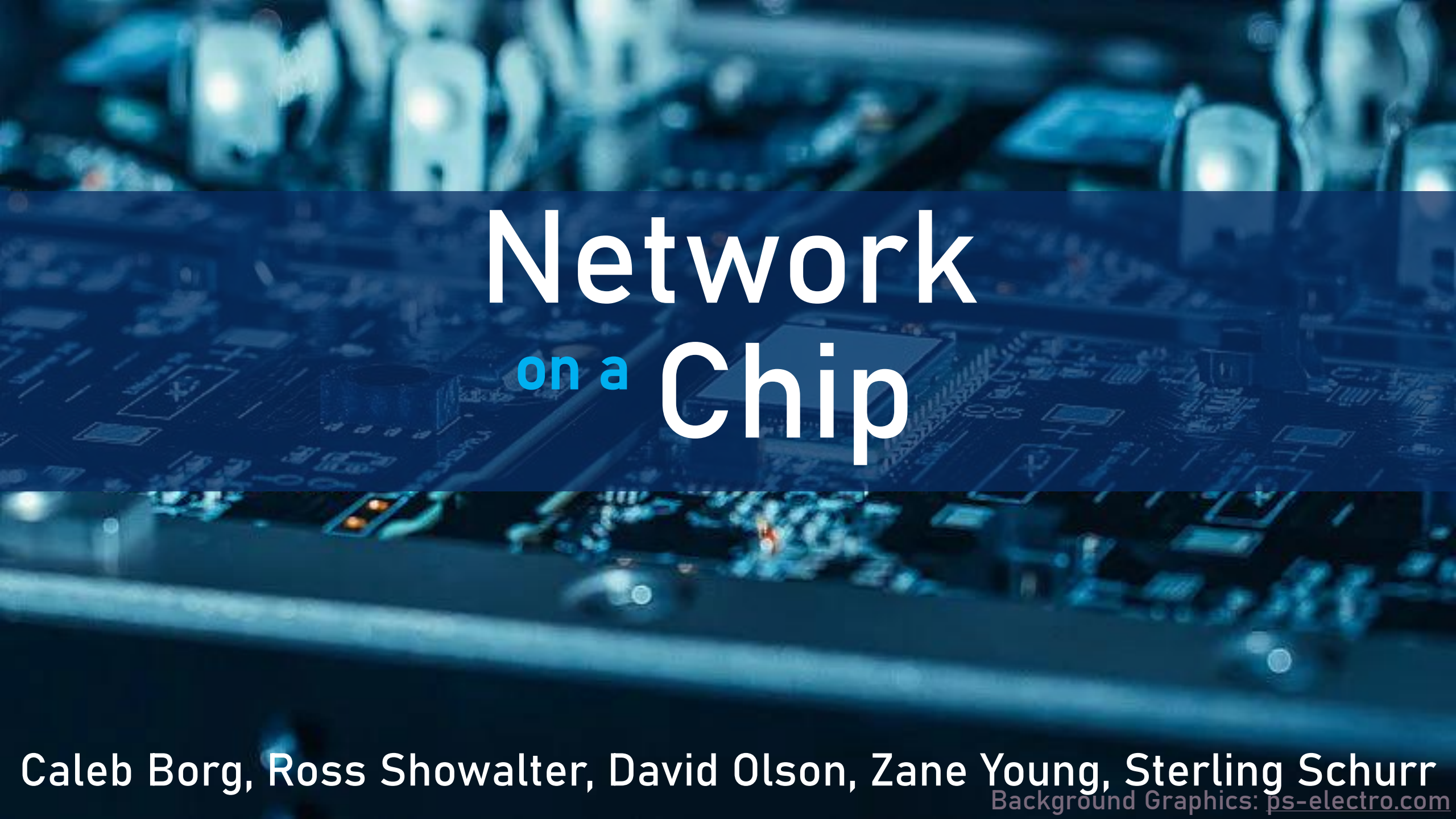
[Model Parallelism](#)

[SambaNova Accelerated Computing with Reconfigurable Dataflow Architecture](#)

[Frontier Journal: CNN Accelerator Dataflow Architecture](#)

[NeuroSpector: Systematic Optimization of Dataflow Scheduling in DNN Accelerators](#)

[Types of Neural Networks](#)

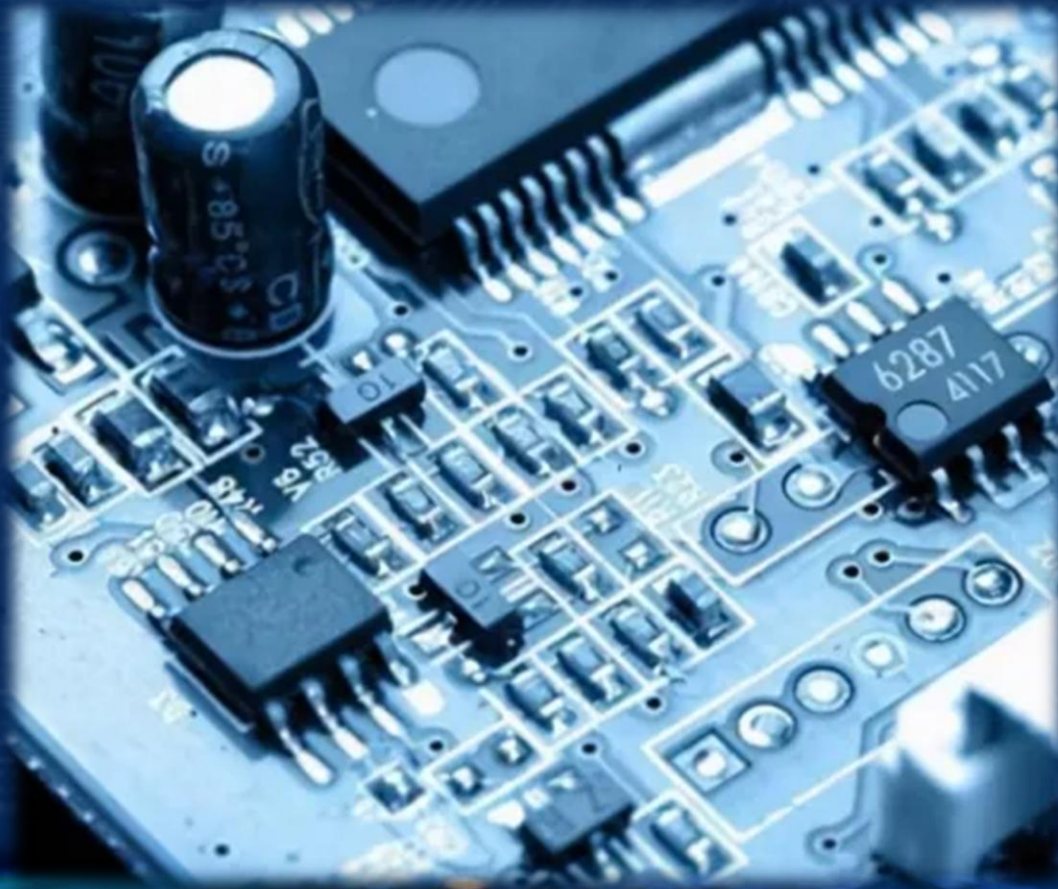


Network on a Chip

Caleb Borg, Ross Showalter, David Olson, Zane Young, Sterling Schurr

Background Graphics: ps-electro.com

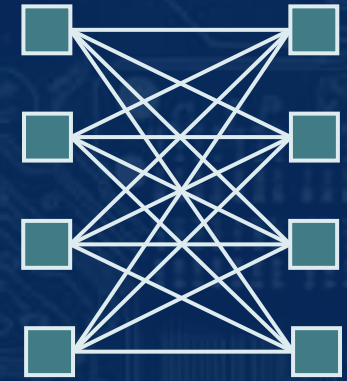
Overview



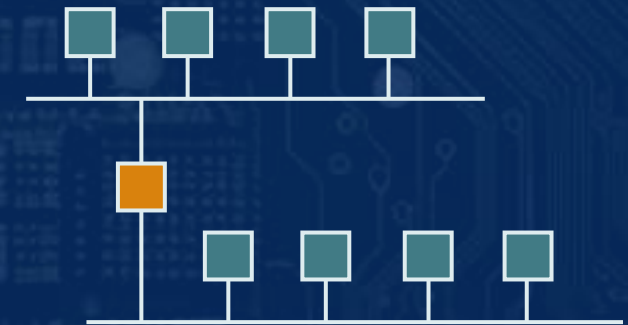
- What is NoC?
- How does it Work?
- Routers
- Flow Control
- Communication Layers and Topology
- Performance and Power
- Real World Examples
- Quiz

What is NoC? Why Do We Use It?

- Like an ethernet network for a system on chip (SoC).
- Connects parts of the chip called **nodes** with paths called **links**.
- Replaces buses or crossbars with a **packet-switched network** to deliver data.
- Benefits: Fewer wires, Scalable, **QoS**.

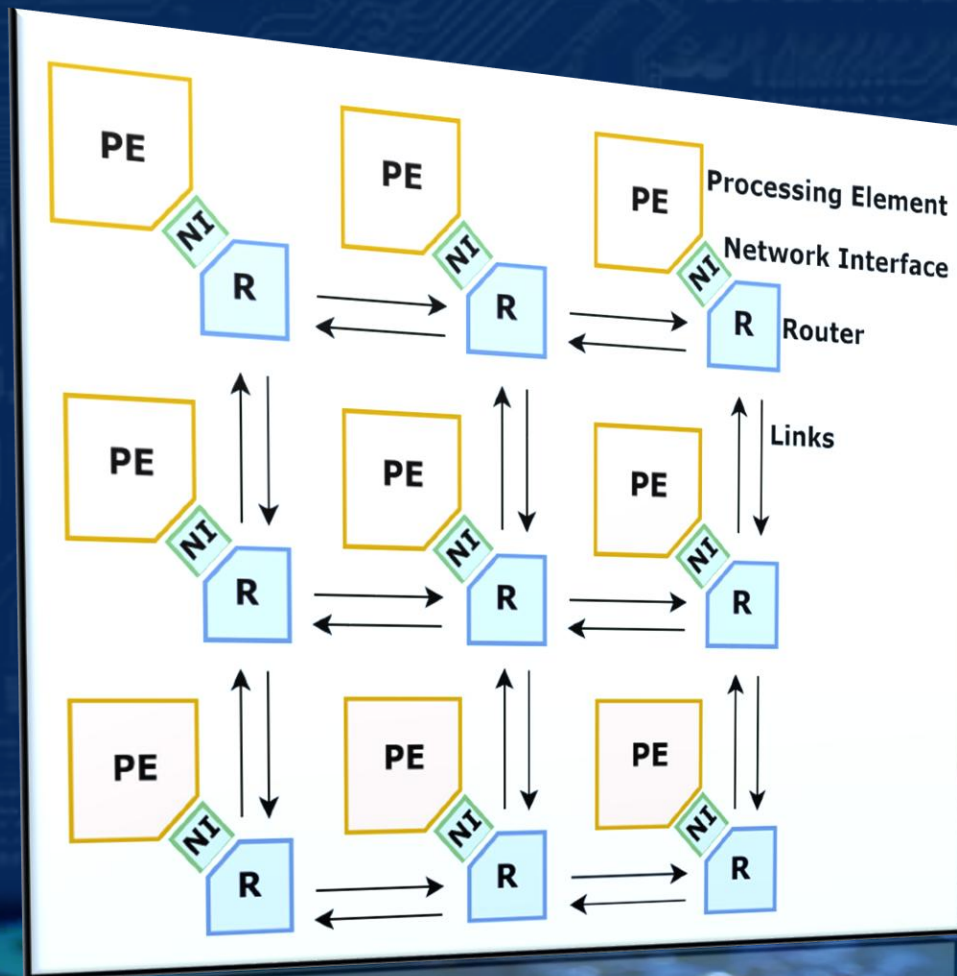


Crossbar



Hierarchical Bus

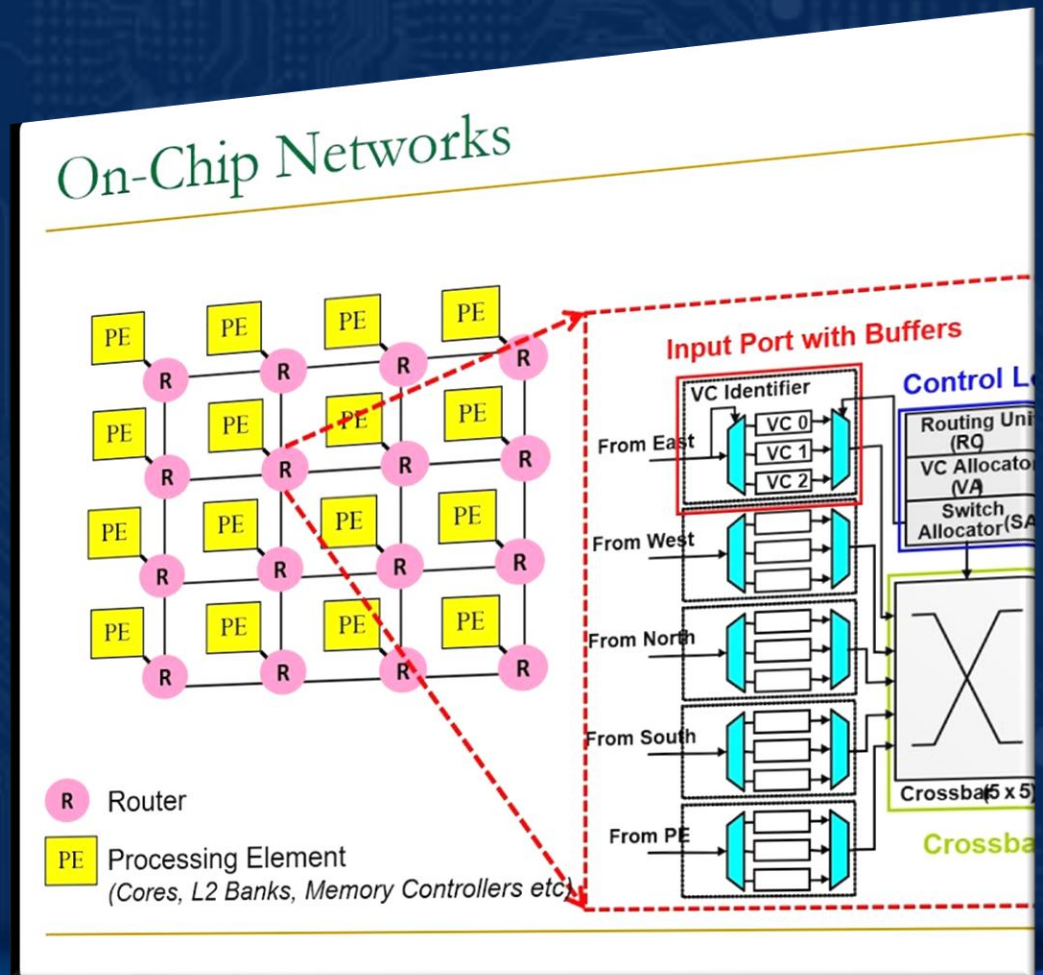
How Does it Work?



- Router based, packet switching.
- Routers determine where to send data and when.
- Network interfaces communicate with devices on chip.

Routers – Finding the best path

- Routers are the logic that selects which output port it should take to reach its destination.
- They usually consist of 5 bidirectional ports (North, South, East, West, and Local Port).
- Core Components consist of I/O ports, Buffers(FIFO), the routing unit, arbitration Unit, and switch matrix.



Routing Classes and Mechanics



Routing Classes

- **Deterministic (static):**
A **pre-determined path**. Can cause congestion between routers.
- **Adaptive (Dynamic):**
Different paths depending on network traffic. Better performance but more complex logic.

Routing Mechanics

- **Source Routing:**
The **sender specifies** the path in the packet header.
- **Distributed Routing:**
Each **router decides** the next hop based on the local routing table.

Flow Control – Managing movement

- Flow Control is the way that routers manage data transmission speeds to prevent buffer over flows and **packet loss**.
- Packets get broken up into smaller groups called **Flits** which allows for less memory to be needed at each router.



Common Techniques



- **Wormhole Routing:** The leading Flit opens a path and then the next **Flits follow like a train**. This is highly effective, but the leading Flit can cause a blocking issue called "**head-of-line blocking**" if it gets stuck.
- **Virtual Channel:** Uses **multiple logical lanes over one physical link** (ethernet) to bypass blocked packets. Unlike wormhole we have lots of virtual lanes allowing flits to go around a stuck flit.
- **Credit-Based Flow Control:** **Routers only send data if the next hop has enough available buffer space**. This gets rid of the clocking issue that can occur as it is only sent if it is told by the receiving router to send.

Communication Layers and Topology



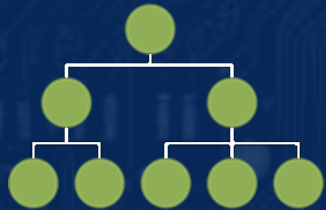
(a) Crossbar



(b) Star



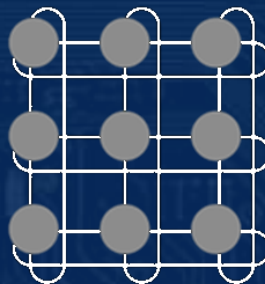
(c) Ring



(d) Tree



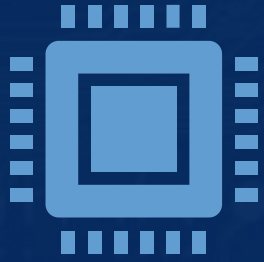
(e) 2D Mesh



(f) 2D Torus

- (a) Crossbar: Matrix connects entire inputs with entire outputs.
- (b) Star: The entire Node connects to a central node.
- (c) Ring: **A single wire connects each node together.**
- (d) Tree: A root node connects to child nodes.
- (e) Mesh: A 4x4 where each router is connected to a compute resource and 4 routers.
- (f) Torus: Like the mesh but there are connections between end nodes.

Performance and Power



How we measure it

Dynamic Power: Network size, components, and usage.

Static power: Regular computing purposes (like transistors).

Efficiency: Often measured in pJ/bit processed.



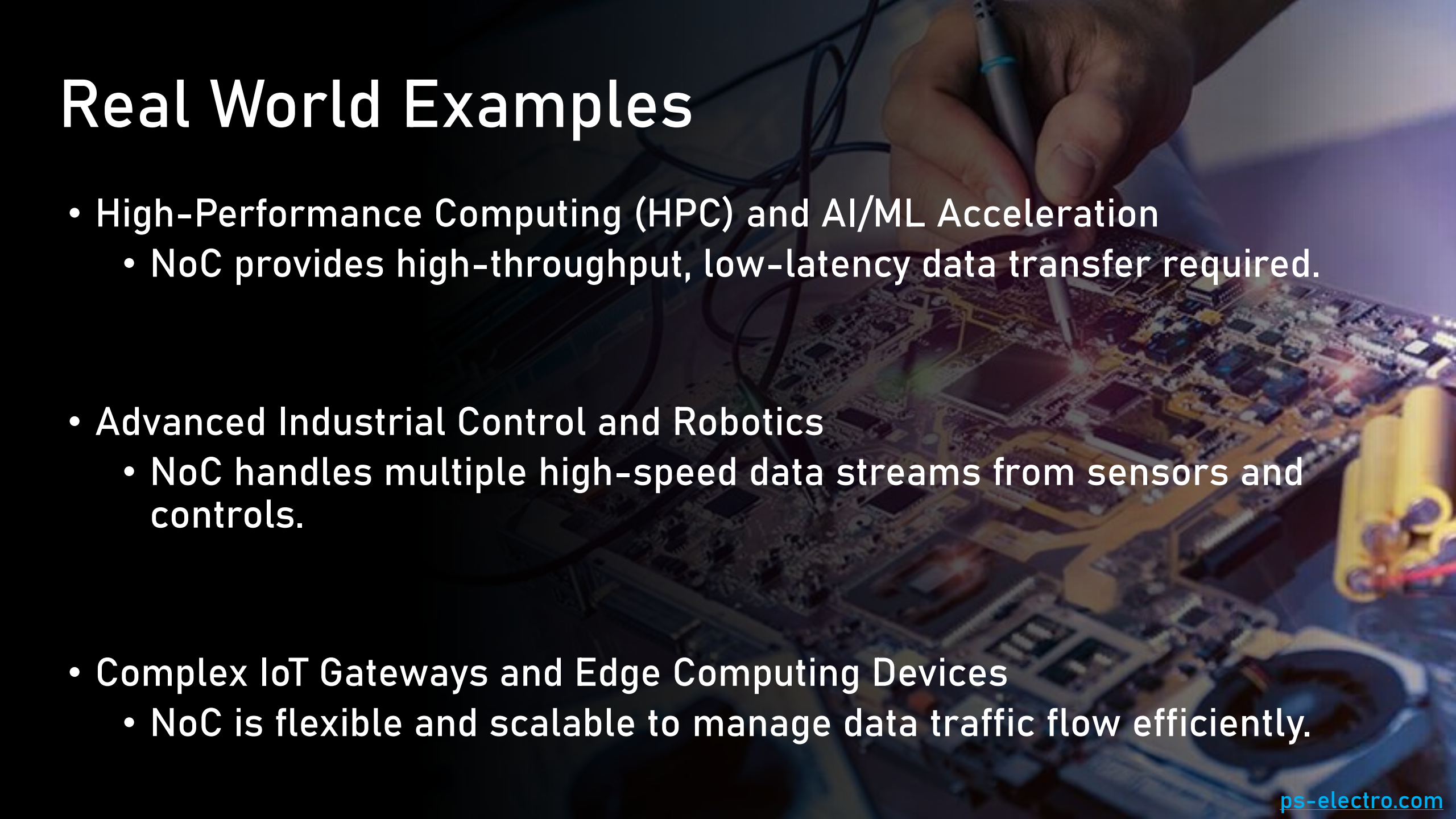
Performance Variability

Performance can vary widely due to many variables.

Network scale, usage, and availability affect throughput.

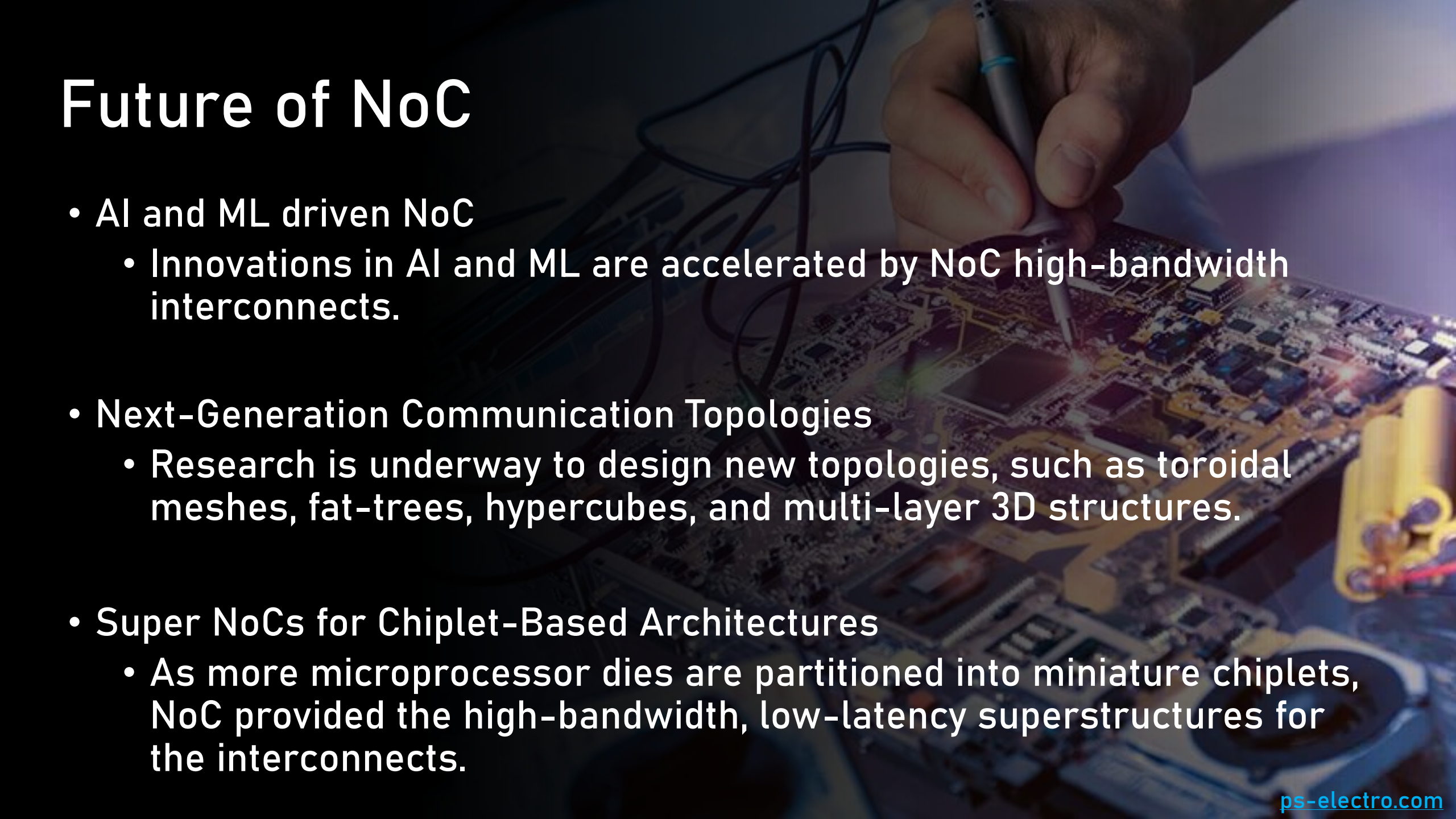
Power usage changes with network usage and application.

Real World Examples

A background image showing a close-up of a person's hand using a soldering iron to work on a complex, multi-layered printed circuit board (PCB). The board is densely packed with various electronic components, including integrated circuits, capacitors, and resistors. The lighting is focused on the work area, highlighting the intricate details of the circuitry and the precision of the soldering process.

- High-Performance Computing (HPC) and AI/ML Acceleration
 - NoC provides high-throughput, low-latency data transfer required.
- Advanced Industrial Control and Robotics
 - NoC handles multiple high-speed data streams from sensors and controls.
- Complex IoT Gateways and Edge Computing Devices
 - NoC is flexible and scalable to manage data traffic flow efficiently.

Future of NoC



- AI and ML driven NoC
 - Innovations in AI and ML are accelerated by NoC high-bandwidth interconnects.
- Next-Generation Communication Topologies
 - Research is underway to design new topologies, such as toroidal meshes, fat-trees, hypercubes, and multi-layer 3D structures.
- Super NoCs for Chiplet-Based Architectures
 - As more microprocessor dies are partitioned into miniature chiplets, NoC provided the high-bandwidth, low-latency superstructures for the interconnects.



Quiz Time

Question #1:

What kind of issue can not be solved by using a Network on Chip instead of a traditional interconnect?

- A) Priority data being blocked by bulk data
- B) High impedance traces between processing elements.
- C) Poorly optimized routing.
- D) Poor scalability.

Question #2:

Which Topology utilizes a single wire that connects each node?

- A) Star.
- B) Mesh.
- C) Ring.
- D) Tree.

Question #3:

How is efficiency in a Network-on-Chip system measured?

- A) Megajoules per gigabyte.
- B) Milliwatts per cycle.
- C) Picojoules per bit.
- D) Milliamps per node.

Question #4:

What are the small groups of packets called?

- A) Flits.
- B) Spitz.
- C) Packies.
- D) Super duper small packet group.

Question #5:

Which of these is a real-world application of NoC?

- A) High-Performance Computing (HPC) and AI/ML Acceleration.
- B) Advanced Industrial Control and Robotics.
- C) Complex IoT Gateways and Edge Computing Devices.
- D) All of the above.

Thank You

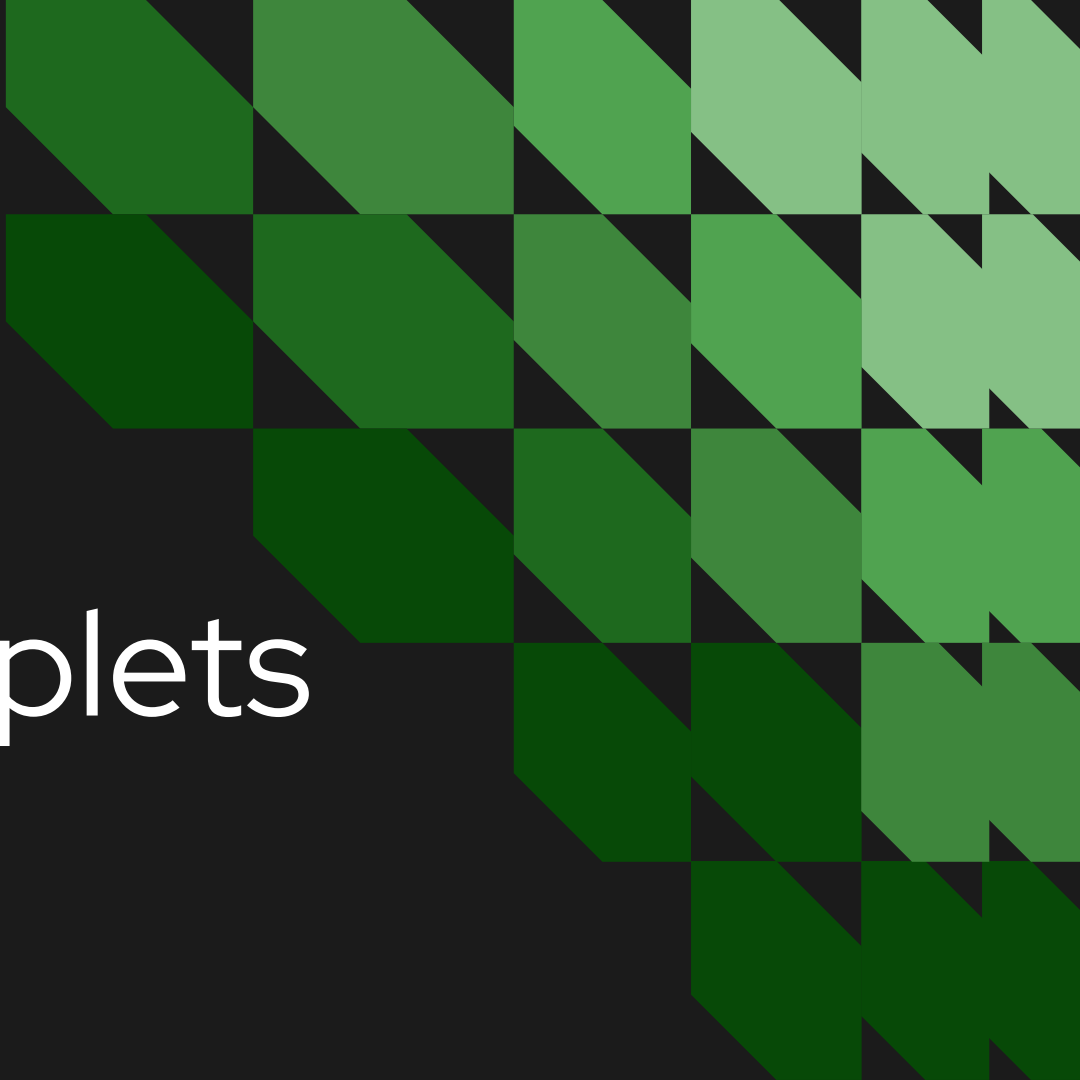


- Sources:

- https://en.wikipedia.org/wiki/Network_on_a_chip
- <https://www.sciencedirect.com/topics/engineering/network-on-chip>
- <https://www.artemis.com/learn/network-on-chip-technology/>
- <https://semiwiki.com/wikis/semiconductor-ip-wikis/network-on-chip-noc-wiki/>
- <https://anysilicon.com/semipedia/network-on-chip-ip-core-noc-ip/>
- <https://semiengineering.com/from-diy-to-advanced-noc-solutions-the-future-of-mcu-design/>
- <https://the-noc-architecture-breakthrough-modern-mcu-design/ps-electro.com> (All Background Graphics)
- [Network-on-Chip Topologies: Potentials, Technical Challenges, Recent Advances and Research Direction | IntechOpen](#)

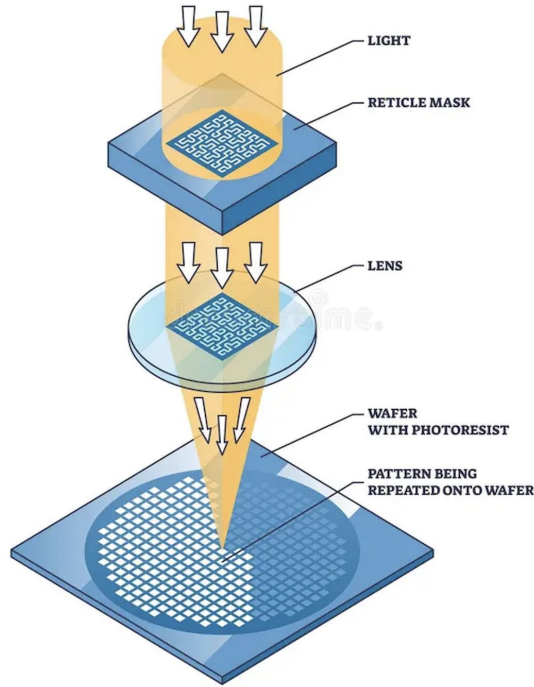
WSE vs. Chiplets

Bryce Langford, Warner Wilkins,
Joel Johnson, Garrison Hammer



The Reticle Limit

EUV LITHOGRAPHY



VectorMine | Dreamstime.com

Photolithography: Projections of circuit patterns onto silicon through a lens: How modern chips are made. (Optical lithography, UV lithography)

Reticle Limit: The maximum area this lens can focus on at once, roughly 26mm x 33mm (858 mm²).

The Problem is AI and High-Performance Computing demand more transistors than can fit in that 858 mm² space.

To get more power, we can no longer just make a bigger single chip using traditional methods. We have to be creative.

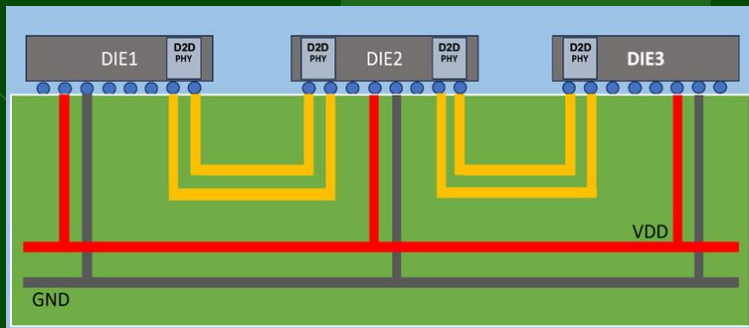
Two Divergent Solutions

The Chiplet Strategy:

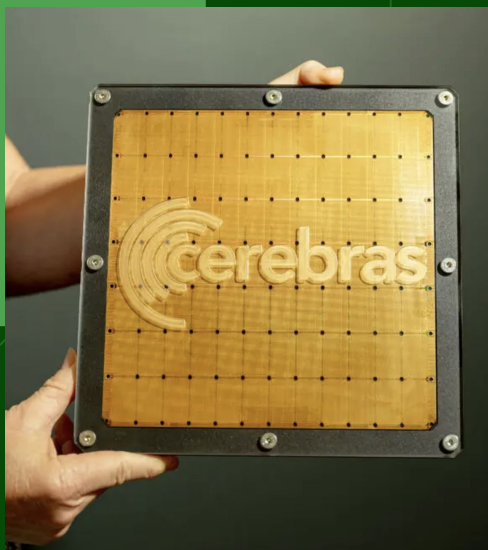
- Used by AMD, Intel, and NVIDIA.
- Method: Break the processor into smaller "chiplets" that are each under the reticle limit.
- These pieces are stitched together on a silicon interposer (a high-speed foundation).

The Cerebras Strategy:

- Used for the Wafer-Scale Engine (WSE-3).
- Method: Instead of dicing the wafer into 500 separate chips, they treat the entire 300mm silicon wafer as one continuous processor.
- They developed cross-reticle wiring to allow data to flow across the lithography "deadzone" boundaries.



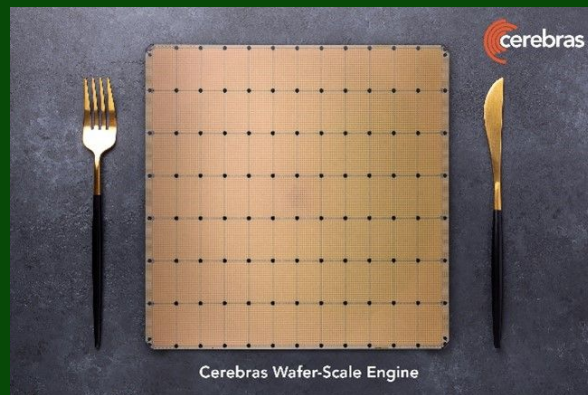
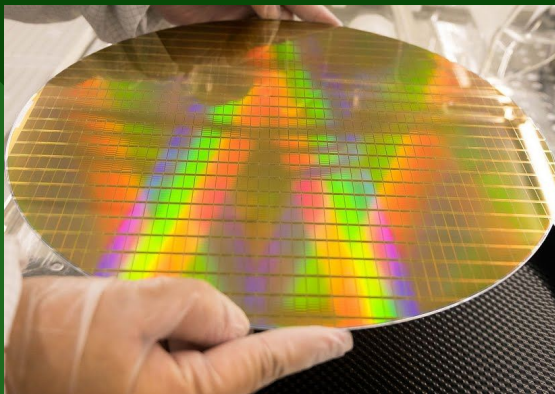
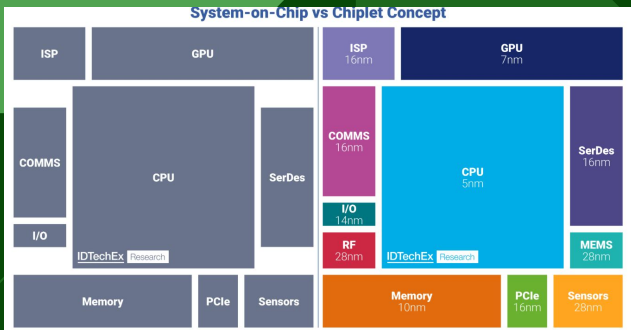
<https://www.keysight.com/blogs/en/tech/sim-des/what-is-a-chiplet-and-why-should-you-care>



Cerebras.com

Manufacturing

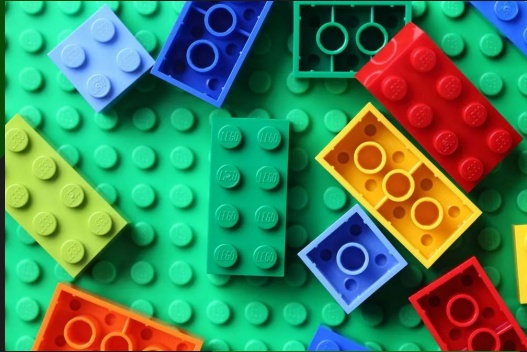
- Chiplets
 - Create many small, specialized chips
 - Test each chip to identify defects
 - Assemble processors from the smaller units
 - Create more functional processors per wafer
- WSE
 - Create one giant chip
 - Include more units than necessary
 - Route around defective areas
 - Create one mega-processor



The Price of Performance

Chiplets, the Economy King

- Chiplets are the winner for being cost effective, if a single 5nm chiplet is bad, the small piece of silicon is tossed.
- Mix-and-Match; older and newer models of chiplets can be combined
- Useful for all applications from laptops to data centers.



Source: (Augustyn, 2024), <https://www.britannica.com/topic/LEGO>

Cerebras, the Speed Demon

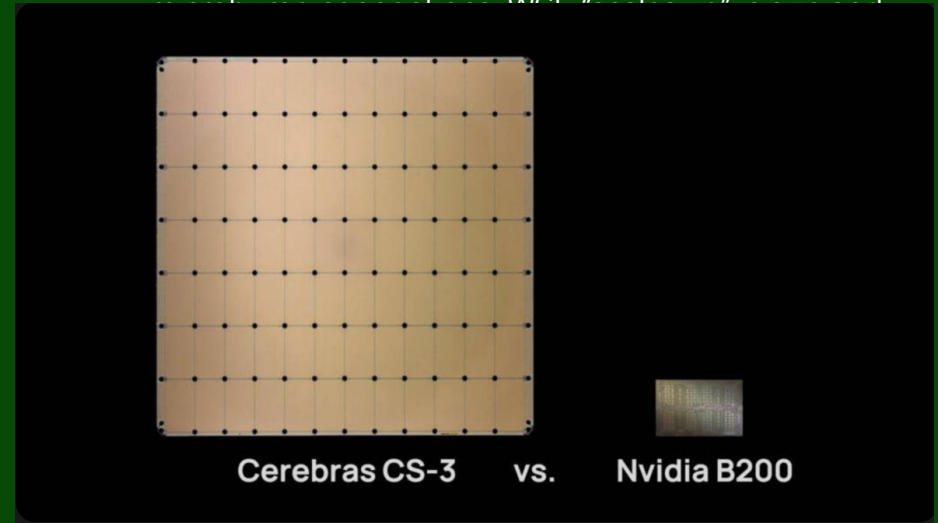
- Massive investment ~3 million dollars per unit
- The CS-3 has 900,000 cores, 4 trillion transistors, with performance of 125 petaflops of AI compute for Specialized infrastructure, requires special cooling and power delivery
- Niche, The best option for LLM training, but too overkill for most other tasks.



Source: (Benson, 2024) <https://www.bbc.com/sport/formula1/articles/c5yrr80x652o>

Scalability

- As more chiplets are added power consumption increases. This is due to data moving between chips.
- In WSE design the entire design is on one piece of silicon, reducing latency by orders of magnitude.
- This means that chiplets “scale out” but can be scaled due to the silicon interposer which has 1024



Source: (McGregor, 2024), <https://futurumgroup.com/insights/cerebras-cs-3-bring-on-the-nvidia-blackwell-competition/>

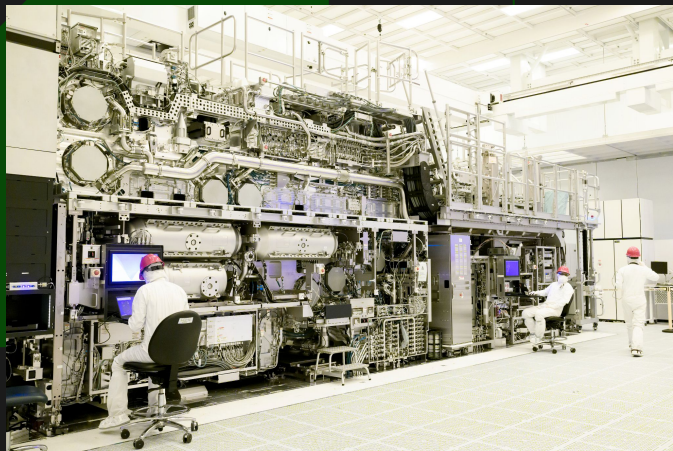
Who Wins the AI Competition?

- The industry is moving towards Universal Chiplet Interconnect Express (UCIe) to make chiplets from various different companies work together.
- Heavy AI computation is likely to be dominated by Wafer-Scale chips and is projected to power the top supercomputers in the near future.
- Overall, both are the future of high performance computing but at different ends of the spectrum. Chiplets for the consumer and many AI uses, while Wafer-Scale for intense computing and pure speed.

Why care?

The Memory Wall

- Processor speed has increased faster than memory access speeds
- Bottleneck is feeding data to the processor



Communication Bottleneck

- GPU clusters have to intercommunicate
- Spend a lot of time idling
- Giant wafer-scale chip increases communication speed by 1000s of times (44GB of SRAM)
- Great for AI workloads

Massive Performance, Minimal Cost.

The CS-3 system is purpose-built for large-scale AI workloads, delivering better performance, lower power consumption, and a smaller footprint compared to GPU-based systems—all at a fraction of the cost.

900,000

AI-Optimized Cores
123x more cores

214PB/s

Interconnect Bandwidth
45,000x more cores

44gb

On-Chip SRAM
1,000x more on-chip memory

21PB/s

Memory Bandwidth
12,800x more bandwidth

1.2Tb/s

System I/O

16RU

System Dimensions

Quiz

1. T/F A WSE is made by creating specialized chips on a wafer and then cutting them out to assemble a custom processor.
2. Why can't engineers simply manufacture a single "monolithic" chip that is 10 times larger than the current industry standard?
 - A) The chips would become too heavy for the circuit boards.
 - B) Current software cannot address more than 1,000 cores at once.
 - C) The Reticle Limit of lithography machines prevents focusing the pattern over a large area.
 - D) Larger chips require more gold, making them too expensive to sell.
3. What is the primary physical technology that makes chiplet architecture possible
 - A) Monolithic Lithography
 - B) Advanced Packaging and Interconnects
 - C) Traditional PCB Copper Traces
 - D) Liquid Nitrogen Cooling
4. T/F GPUs training AI spend significant amount of their time idling.

Reading List/Sources

Cerebras Whitepapers:

<https://www.cerebras.ai/whitepapers>

Cerebras Collaboration:

https://www.sandia.gov/research/news/cerebras-collaborations-cs3/?utm_source=chatgpt.com

Lithography Principles:

<https://www.asml.com/en/technology/lithography-principles/rayleigh-criterion>

Chiplets:

<https://semiengineering.com/outlook-2025-embracing-chiplets/>
<https://www.idtechex.com/en/research-article/chiplets-revolutionizing-semiconductor-design-and-manufacturing/32101>
<https://www.keysight.com/blogs/en/tech/sim-des/what-is-a-chiplet-and-why-should-you-care>

Amd instinct:

<https://www.amd.com/content/dam/amd/en/documents/instinct-tech-docs/data-sheets/amd-instinct-mi300x-data-sheet.pdf>

Ai workloads:

<https://www.cudocompute.com/blog/select-gpu-for-ai-workloads>

Arxiv.org:

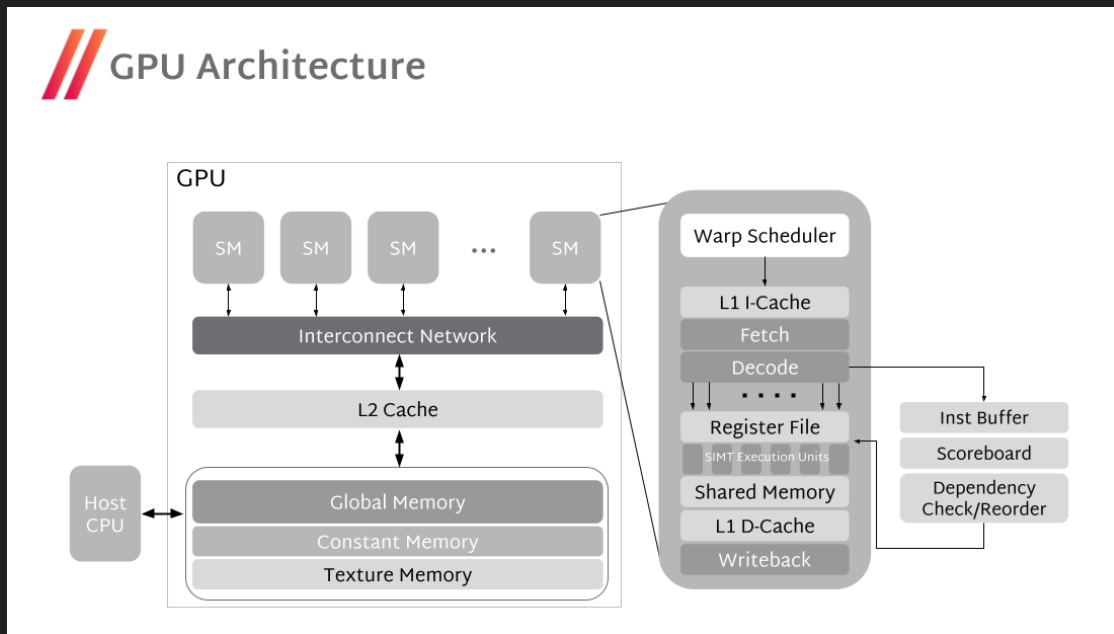
<https://arxiv.org/html/2503.11698v1>

GPUs

Samuel, Tyler, Bryce, Connor

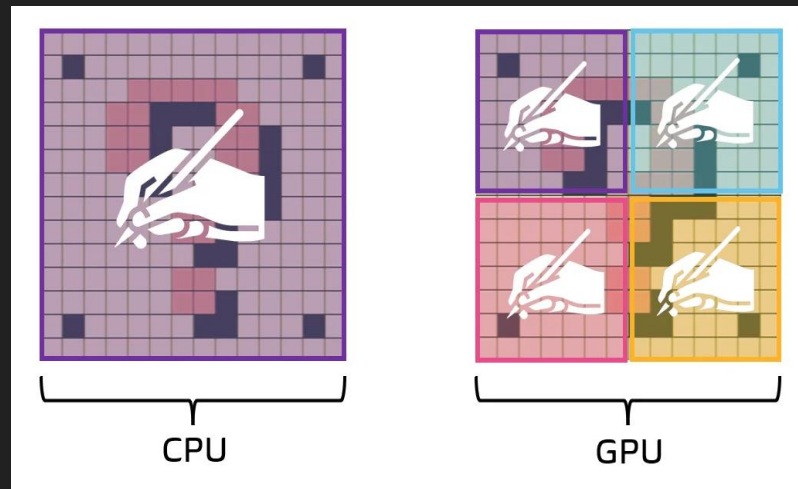
GPU Architecture Fundamentals

- GPUs specialize in parallelism
- Thousands of small cores
 - CUDA cores for NVIDIA
 - Stream processors for AMD
- Memory
 - L1 Cache
 - L2 Cache
 - VRAM
 - High bandwidth memory (HBM)
- SIMD/SIMT



The GPU Programming Model

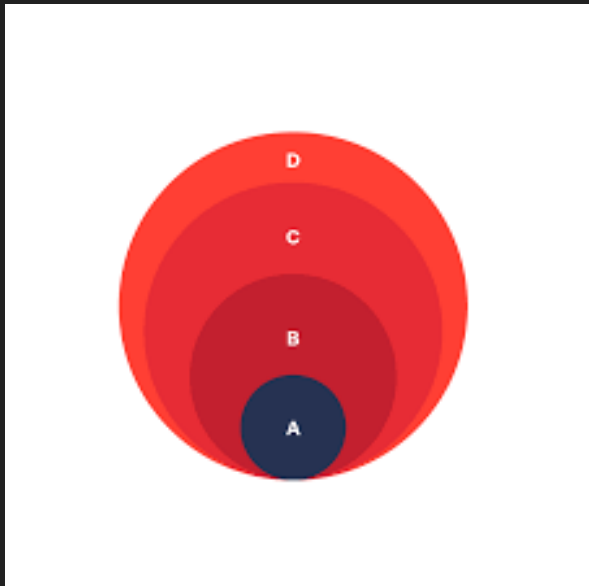
- The Mental Model Shift
 - Latency vs. Throughput:
 - CPUs "Fastest to Finish" (Serial optimization)
 - GPUs "Most Finished at Once" (Parallel optimization)
 - The Host/Device Relationship:
 - CPU (Host) is the manager handling logic & OS tasks
 - GPU (Device) is the specialized worker for data-crunching
 - The Offload Pattern:
 - Software is "sent" by the CPU
 - Data must be copied to the Device before work begins



CPU vs GPU -

https://www.reddit.com/r/ElectricalEngineering/comments/1bo98g6/cpu_vs_gpu/

The GPU Programming Model



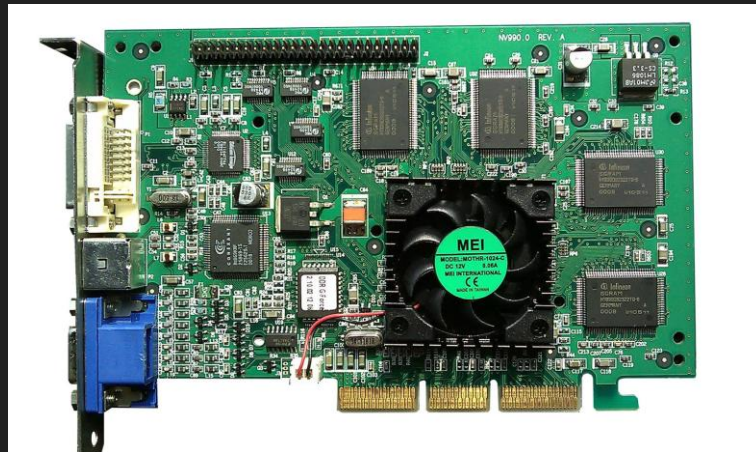
D: The Grid
C: The Block
B: The Warp (Group of 32 threads)
A: The Thread

- The Execution Hierarchy

- The Kernel:
 - The Blueprint - C-style function for every parallel thread
 - Massive Parallelism - One kernel executed by thousands of threads
- The Software Hierarchy:
 - Threads are the smallest unit of work (one pixel/value)
 - Blocks are groups of threads sharing local memory
 - Grids are the total problem space launched by the Host
- The Hardware Mapping
 - Dispatch - CPU launches the Grid onto the GPU
 - Scheduling - GPU hardware maps Blocks to physical cores

GPUs as General-Purpose Accelerators (GPGPU)

- The original GPU
 - Used exclusively for graphics
 - Rasterization pipelines
 - Vector shading
 - Texture mapping
 - Just told pixels what to do

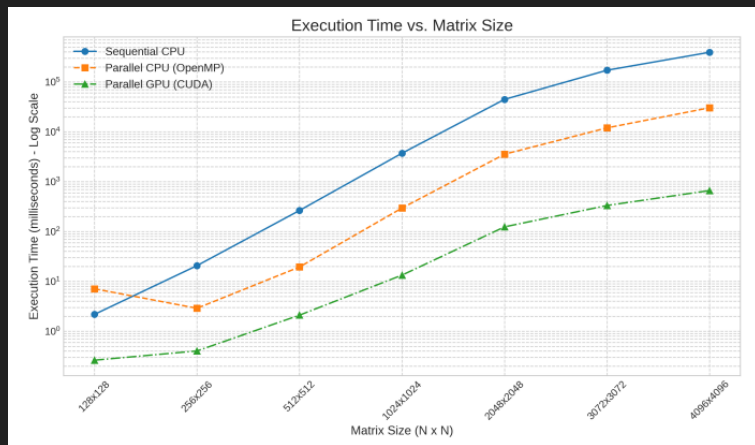


Hyins (2010), NVIDIA GeForce 256,
https://commons.wikimedia.org/wiki/File:VisionTek_GeForce_256.jpg

- Pixels are data
 - The floating point units could be used to compute *anything* numerically
 - Computer scientists were encoding their data as textures to trick the GPU into doing math

GPUs as General-Purpose Accelerators (GPGPU)

- Why GPUs beat CPUs at parallel workloads
 - CPUs optimize for fast sequential execution; GPUs optimize for massive parallel throughput
 - CPUs have several cores; GPUs have thousands of smaller cores
 - GPUs have enormously wide memory busses
 - GPUs win out when the same operation is being applied to huge datasets.
- Examples:
 - Matrix multiplication
 - Statistics
 - Linear Algebra
 - Convolution
 - FFT



Ansari et al. (2025),
"Accelerating Matrix
Multiplication: A
Performance
Comparison Between
Multi-Core CPU and
GPU"

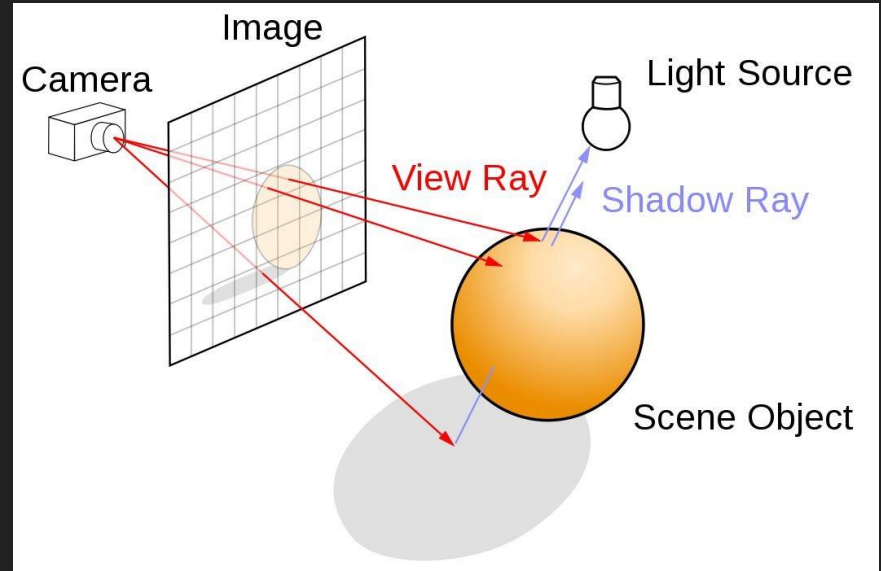
Modern GPU Applications

GPUs are indispensable in:

- Ray Tracing
- Generative AI
- Calculations in things like:
 - Medicine
 - Weather
 - Automation

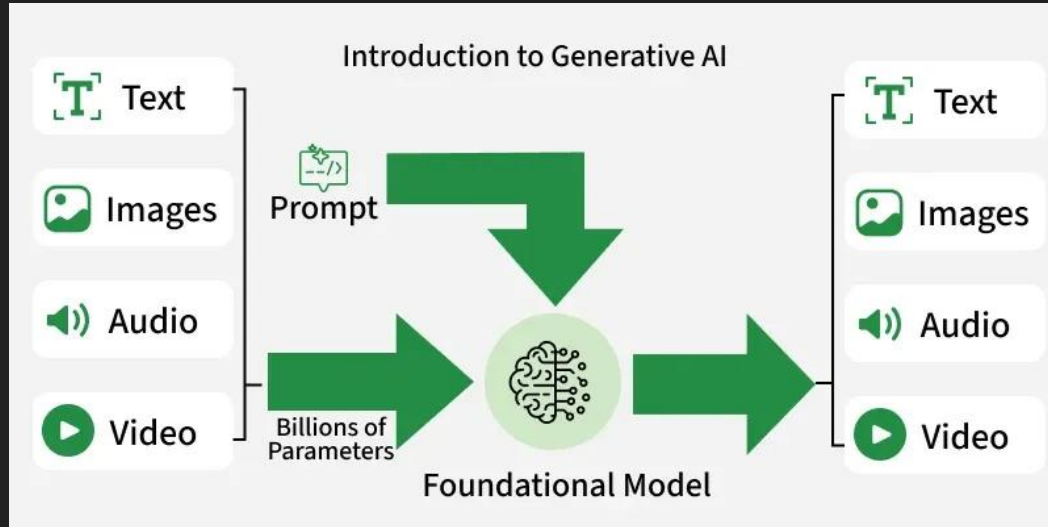
Ray Tracing

- Rendering technique that can realistically simulate the lighting of a scene
 - Ray Casting
 - Traces ray from camera through each pixel in an image plane
 - Path Tracing
 - More intense ray tracing (hundreds of thousands of rays through each pixel)
 - Rasterization
 - Displays 3D objects on a 2D screen
- Very computationally heavy, takes CPU based farms multiple days to render
- Specialized cores, such as Tensor cores in NVIDIA GPUs, handle the complex math at high speeds



<https://developer.nvidia.com/discover/ray-tracing>

Generative AI



<https://www.geeksforgeeks.org/artificial-intelligence/what-is-generative-ai/>

- GPUs enable training and deployment of complex AI models
 - High throughput
 - Parallelism
- Training AI
 - Lots of complex math operations that adjust model parameters
 - Feed the model lots of data
 - Accelerated by simultaneous operations via GPUs
- Running AI
 - Real time predictions on new data
 - Rapid calculations to make predictions at high speeds, enabled through GPUs

Questions

1. What is the main difference between a CPU and a GPU?
 - a) GPUs have hundreds of cores and are made for parallelism
 - b) CPUs have cache while GPUs don't
 - c) GPUs lack the ability to do floating point operations
 - d) There is no difference between a CPU and a GPU
2. If a GPU is executing a "Grid" of 10,000 threads, how many different "Kernels" are typically being run?
 - a) 10,000 (One unique kernel per thread)
 - b) 312 (One kernel per Warp)
 - c) 0 (Kernels only stay on the CPU)
 - d) 1 (One blueprint executed by all threads)
3. What was the primary barrier to GPGPU computing before NVIDIA released CUDA in 2007?
 - a) GPUs didn't have enough memory to store large datasets
 - b) Programmers had to encode data as textures to trick the GPU into doing math
 - c) GPUs could only perform integer arithmetic, not floating-point
 - d) There was no way to transfer data between CPU and GPU memory
4. In the context of GPU acceleration, which of the following best describes why specialized hardware (i.e Tensor/CUDA Cores) is critical for both Ray Tracing and Generative AI?
 - a) They allow the GPU to switch between tasks like a CPU
 - b) They handle the massive volume of simultaneous math operations
 - c) They are used to decrease the "high throughput"
 - d) They replace the need for path tracing by allowing a single ray to be traced through every pixel

Articles

“ACCELERATING MATRIX MULTIPLICATION: A PERFORMANCE COMPARISON BETWEEN MULTI-CORE CPU AND GPU” - <https://arxiv.org/pdf/2507.19723>

“General-purpose computing on graphics processing units” - https://en.wikipedia.org/wiki/General-purpose_computing_on_graphics_processing_units

“2678x Faster Matrix Multiplication with a GPU” - <https://0mean1sigma.com/2678x-faster-how-gpus-supercharge-matrix-multiplication/>

“A brief history of GPU” - <https://medium.com/altumea/a-brief-history-of-gpu-47d98d6a0f8a>

“A Survey of General-Purpose Computation on Graphics Hardware” - https://research.nvidia.com/sites/default/files/pubs/2005-08_A-Survey-of/ASurveyofGeneralPurposeComputationonGraphicsHardware.pdf

“CUDA C++ Programming Guide” - <https://docs.nvidia.com/cuda/cuda-c-programming-guide/>

“The GPU Computing Era” - https://pages.cs.wisc.edu/~markhill/restricted/ieeemicro10_gpu.pdf

“Teaching Accelerated CUDA Programming with GPUs” - <https://developer.nvidia.com/how-to-teach-cuda>

“Ray Tracing” - <https://developer.nvidia.com/discover/ray-tracing>

“What is a GPU & Its Importance for AI” - <https://cloud.google.com/discover/gpu-for-ai>

“What is Generative AI” - <https://www.geeksforgeeks.org/artificial-intelligence/what-is-generative-ai/>

Articles Continued

“Tutorial: Understanding GPU Architecture and Execution Model” <https://eunomia.dev/others/cuda-tutorial/04-gpu-architecture/>

“A Gentle Introduction to GPU Programming” - https://bssw.io/blog_posts/a-gentle-introduction-to-gpu-programming

“CUDA Refresher: The CUDA Programming Model” - <https://developer.nvidia.com/blog/cuda-refresher-cuda-programming-model/>

In-Memory Computing

Seth Ricks, Ian McMaster, Emily Sammons, Jonas Norwood

Overview

What is In-memory computing?

What is the problem it is solving?

Methods used

Applications

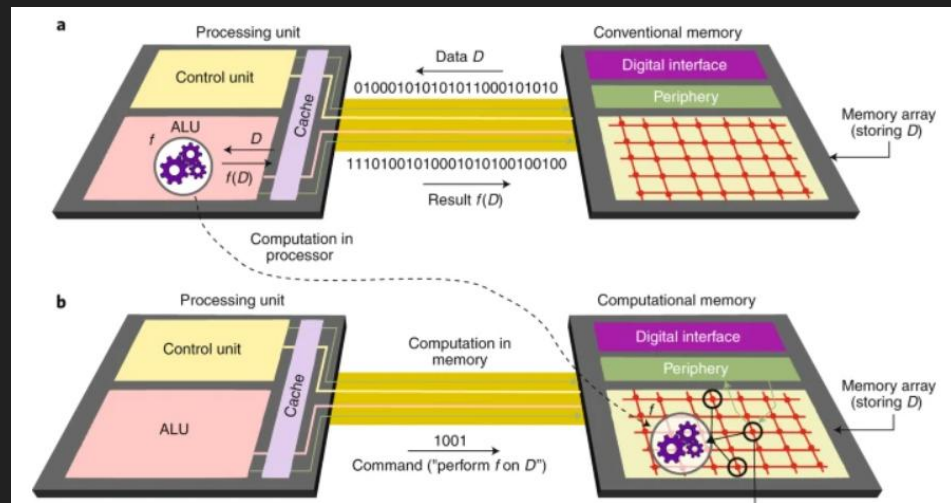
Challenges

Test questions

What is In-Memory Computing?

“IMC ... [is] where memory is not only storing data but can also directly output computation results.”
(Paper from Association of Computer Machinery)

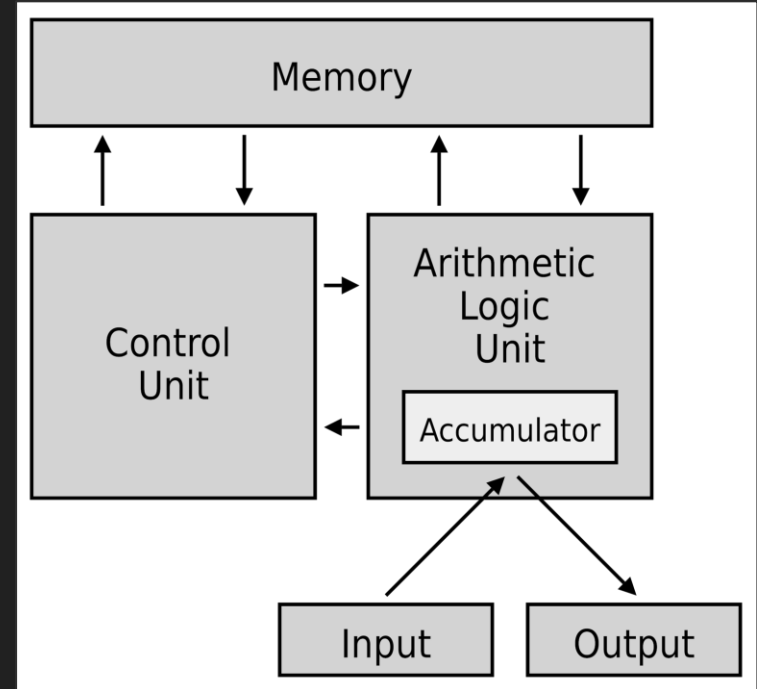
- Perform computations directly in memory instead of just storing information.
- Addressing “memory wall” (CPU has grown much faster than memory speeds)
- Helps save energy with less data movement.



Sebastian et al., Nature Nanotechnology, 2020

What is the problem is it solving?

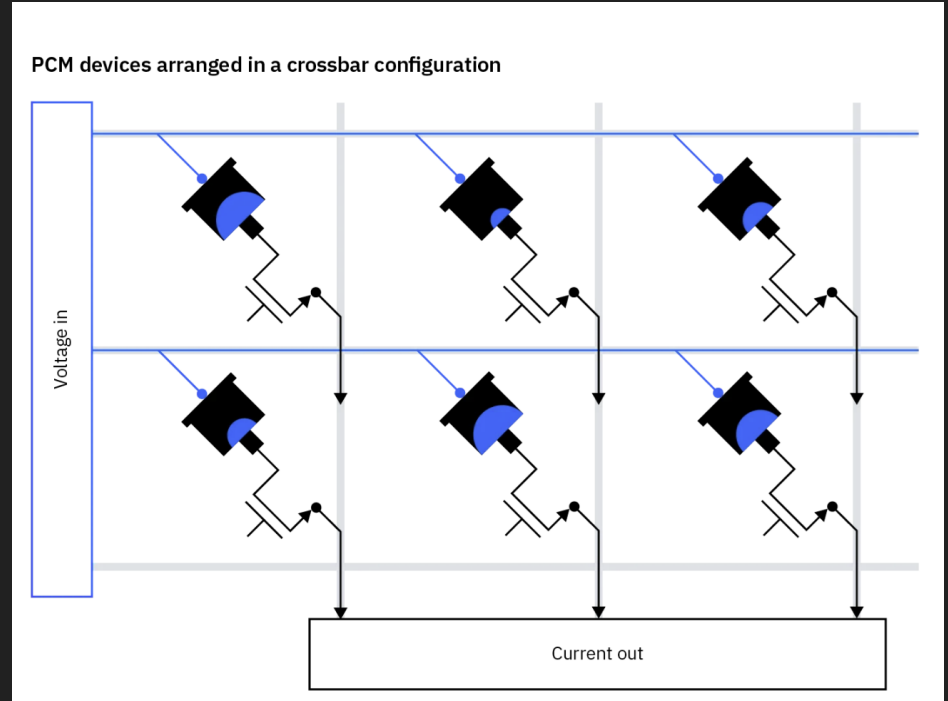
- The Von-Neumann computer model relies heavily on memory.
- Passing data to and from memory is power and time demanding.
- Cpu's have improved a lot, and memory has improved less.
- AI has made this more apparent as the AI models use lots of memory.



(Figure from Digikey)

PCM Devices - A Method of In-Memory Computing

- PCM devices use Ohm's and Kirchhoff's Laws to do matrix multiplication.
- “Moving 64 bits of data from DRAM to CPU consumes 1-2nJ, which is 10,000–2,000,000 times more energy than is dissipated in a PCM device performing a multiplication operation (1-100fJ).” (Le Gallo-Bourdeau)



(Figure from IBM)

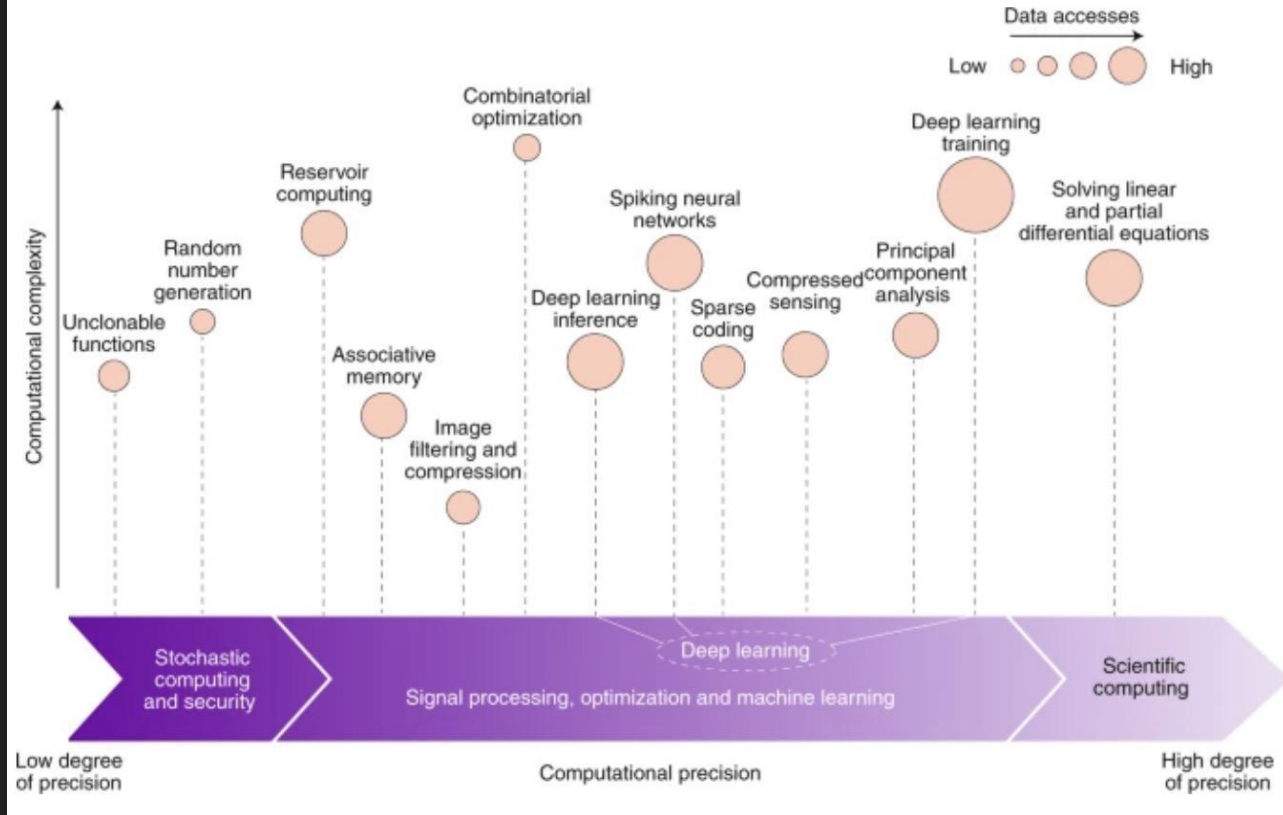
Potential Applications

- Neural Network Computations (AI)
 - AI processes large amounts of data
 - IMC would decrease computation time
- Signal Processing
 - Complex signal chains could travel through memory
- Stochastic Computing*
 - Benefits from data being closer to computation
- Security (Cryptography, hashing, etc.)
- Solving Differential Equations

*Probability based computing

(All computation can gain efficiency since there would be less memory transfers)

Fig. 4: The application landscape for in-memory computing.



Challenges of In Memory Computing

- Integration

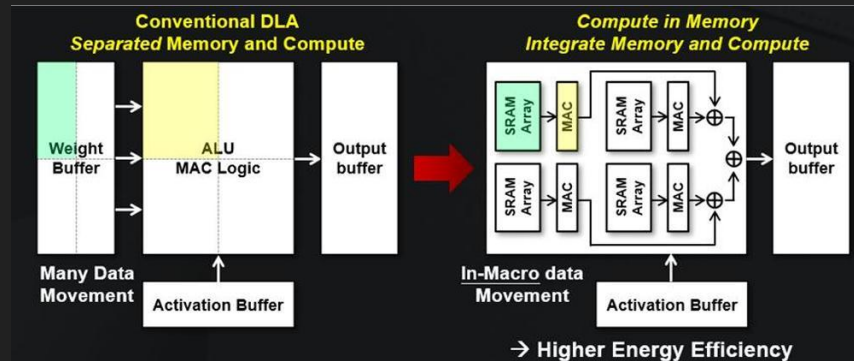
- Memory chips are optimized for density instead of logic like previous models.
- Coding is optimized for Von - Neumann architecture,
- Existing systems (SAP HANA)

- Functions

- Currently not a full CPU replacement (simple arithmetic vs floating point)
- Data consistency and Cache coherence
- Precision of analog computing

- New, faster memory is expensive to manufacture, ReRam, MRam, etc.

- PIM introduces new side channels for attacks



Research

- Overview
 - <https://www.nature.com/articles/s41565-020-0655-z>
- Currently being researched for AI applications:
 - [A Heterogeneous and Programmable Compute-In-Memory Accelerator Architecture for Analog-AI Using Dense 2-D Mesh | IEEE Journals & Magazine | IEEE Xplore](#)
 - [Using the IBM analog in-memory hardware acceleration kit for neural network training and inference | APL Machine Learning | AIP Publishing](#)
 - [Hardware-aware training for large-scale and diverse deep learning inference workloads using in-memory computing-based accelerators | Nature Communications](#)
 - [A 64-core mixed-signal in-memory compute chip based on phase-change memory for deep neural network inference | Nature Electronics](#)
 - [In-memory factorization of holographic perceptual representations | Nature Nanotechnology](#)
- In Memory Computation With Phase Change Memory:
 - [Projected Mushroom Type Phase-Change Memory - Ghazi Sarwat - 2021 - Advanced Functional Materials - Wiley Online Library](#)

Test Questions

1. Why is In-Memory Computing being explored?

- a. To reduce the number of transistors used in memory.
- b. To reduce the time and power spent moving data between the CPU and memory.
- c. To save space on the motherboard by removing cache.
- d. To make register-to-register transfers faster.

2.) True / False

DRAM to CPU transfers can consume 10,000–2,000,000 times more energy than what is dissipated by other solutions doing the matrix multiplication operation in memory.

Test Questions

3.) True / False

Memory has improved at the same rate as CPU's.

4.) Select All That Could Benefit From In Memory Computing:

- Artificial Intelligence
- Signal Processing
- Low Memory Applications
- Solving Differential Equations
- None of the above

Citations

- Y. Ma, Y. Du, L. Du, J. Lin, and Z. Wang, “In-memory computing: The next-generation AI computing paradigm,” in *Proc. ACM Great Lakes Symp. VLSI (GLSVLSI '20)*, Virtual Event, China, Sep. 2020. doi: 10.1145/3386263.3407588.
- Tian, P. H., Yousefijamarani, Z., & Alameldeen, A. (2025, October 9). *DL-PIM: Improving data locality in processing-in-memory systems*. arXiv.org. https://arxiv.org/abs/2510.07719?utm_
- Kaur, R., Asad, A., & Mohammadi, F. (2024, July 16). *A comprehensive review of processing-in-memory architectures for deep neural networks*. MDPI. <https://www.mdpi.com/2073-431X/13/7/174>
- Chen, W., Qi, Z., Akhtar, Z., & Siddique, K. (2022, November 9). *Resistive-RAM-based in-memory computing for neural network: A Review*. MDPI. <https://www.mdpi.com/2079-9292/11/22/3667>

Additional Citations

- M. Le Gallo, “The hardware behind analog AI,” IBM Research Blog, Nov. 20, 2021. [Online]. Available: <https://research.ibm.com/blog/the-hardware-behind-analog-ai>
- IBM Technology, “What is In-Memory Computing?,” YouTube, [Online]. Available: <https://www.youtube.com/watch?v=BTnr8z-ePR4>
- Asianometry, “AI’s Hardware Problem,” YouTube, [Online]. Available: <https://www.youtube.com/watch?v=5tmGKTNW8DQ>
- Veritasium, “Future Computers Will Be Radically Different (Analog Computing),” YouTube, [Online]. Available: <https://www.youtube.com/watch?v=GVsUOuSjvcg>
- Forbes, “Efficient AI Inference With Analog Processing In Memory,” YouTube, [Online]. Available: <https://www.youtube.com/watch?v=b1F1h1U3vvs>



FPGAs



FPGA architecture and how they are
used as accelerators

Adam Sandland, Taylor Alvey, Jeffrey Fuhriman, Sam Cook



What Is an FPGA and What Does It Do?

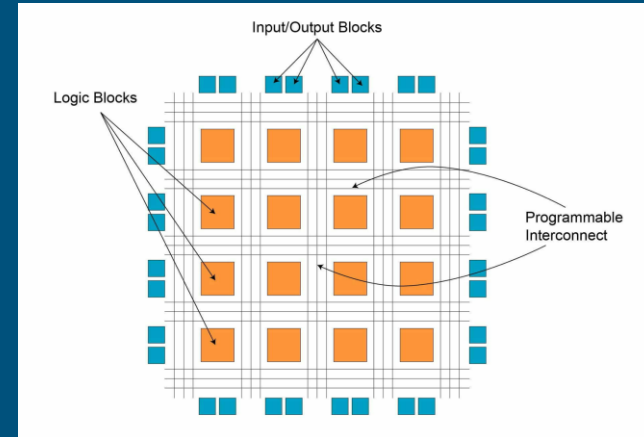
- FPGA stands for Field-Programmable Gate Array.
- It is a chip that can be reconfigured even after manufacturing.
- Instead of only running software, an FPGA can be configured to act like custom digital hardware.
- FPGAs are used to accelerate specific tasks by running many operations in parallel.



Source: Niranjana R, Logic Fruit Technologies

FPGA Architecture

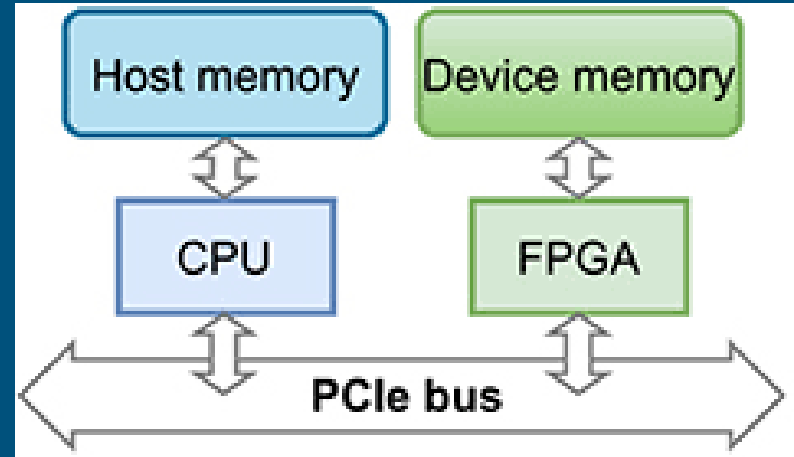
- Built from many small configurable logic blocks.
- Lookup tables (LUTs) implement combinational logic.
- Registers/flip-flops store data for sequential logic.
- Programmable interconnects connect blocks together.
- I/O blocks communicate with external devices.
- Many FPGAs also include Block RAM and DSP slices.



Source: Dheeraj Punia, Logic Fruit Technologies

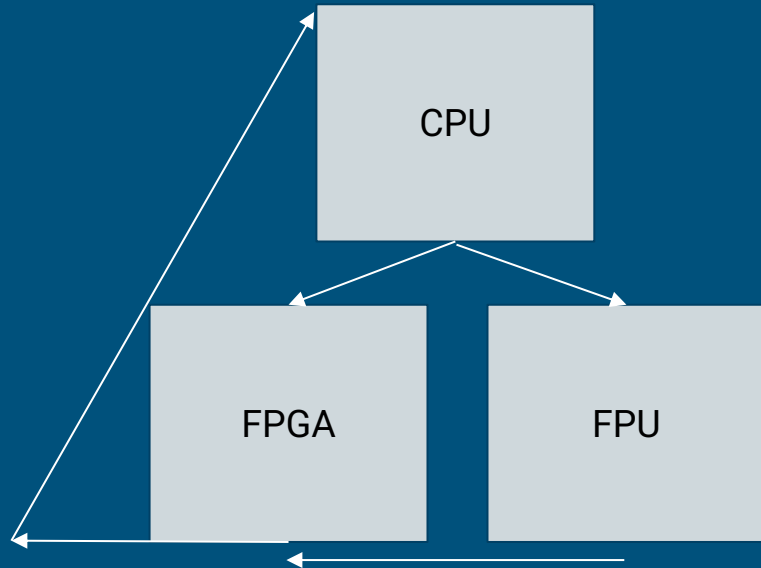
What do FPGAs do for CPUs?

1. Most FPGAs are off-cpu chips
2. They can be programmed by the CPU
3. Can provide throughput for the CPU



How do we speed up processes with FPGAs?

- CPUs can push complex processes to FPGAs
 - This can make effective hardware-optimized functions for certain code or pipelines
- Hardware-baked parallelism
- Negligible clock cycles for operations
- Lower Power consumption than CPUs



Common Applications of FPGA Accelerators

AI / ML

$$y = w_1x_1 + w_2x_2 + w_3x_3 + \dots + w_nx_n$$

- Custom parallel multiply-accumulate (MAC) operations (100s-1000s at once)
- Speeds up & reduces latency of neural networks doing matrix multiplication

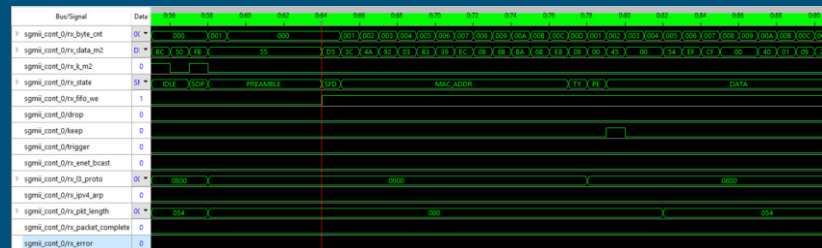
Data Centers & Cloud Computing

- Encryption runs in parallel for VPN, HTTPS servers, cloud storage
- Hardware pipelines for compression (video streaming, cloud storage)
- Faster large database searching using parallel data comparisons



Networking & Signal Processing

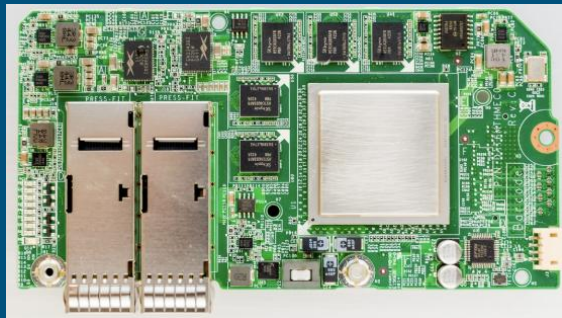
- Hardware packet processing (no CPU delay)
- Routers, 5G, radar, radio
- Parallel logic allows high throughput speeds and low latency



Project Catapult



- Starting in 2010, started building the world's first AI supercomputer
- Doug Burger, James Larus, & Andrew Putnam revolutionized FPGAs to solve “the eventual end of Moore’s Law”
- FPGAs became the “front door”, making decisions without the processor’s involvement
- By 2015, ran decision tree algorithms 40x faster than CPUs alone



FPGA vs CPU

- FPGA handles compute-intensive, deeply pipelined, massively parallel operations. CPU handles the rest.
- CPU is fixed instruction set and rigid memory hierarchy. FPGA is a flexible, fully customizable architecture.
- CPU developer adapts the program to the architecture. FPGA developer adapts the architecture to the program.

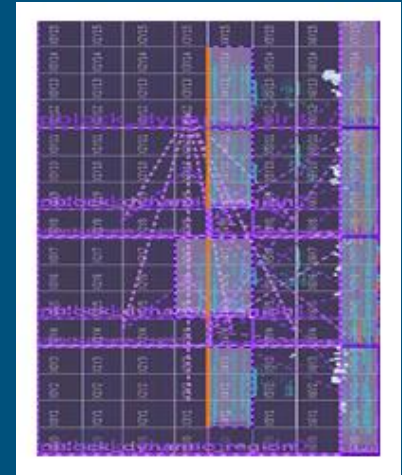
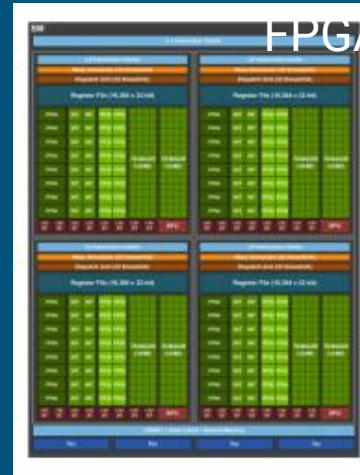
Without acceleration



With FPGA acceleration



CPU



(Source: Xilinx)

FPGA Customization

What is customizable?

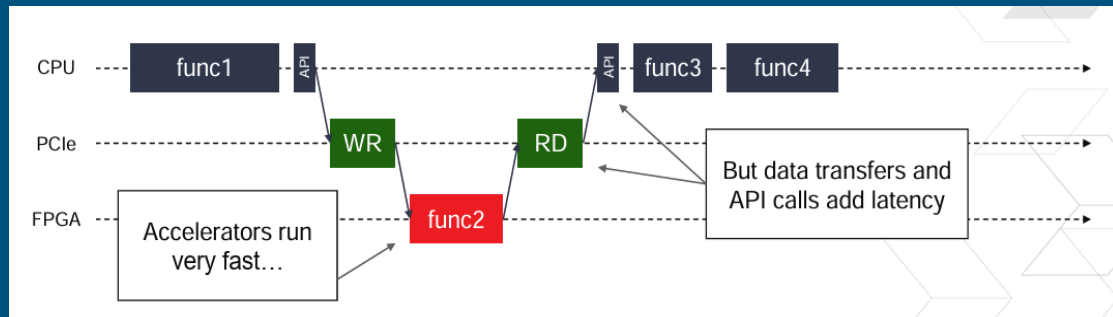
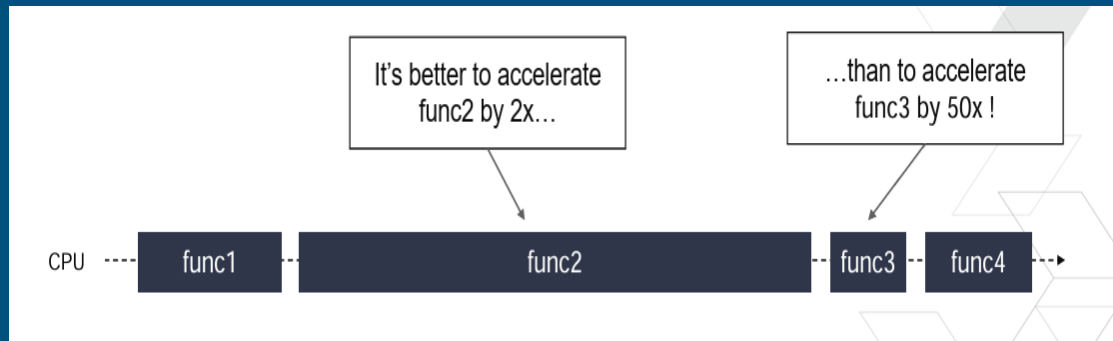
- Custom datapaths
- Custom bit-width
- Custom memory hierarchies

What it is good at?

- Deeply pipelined
- Bit manipulations
- Wide datapath
- Custom memory hierarchy

Why it does a good job?

Adapts to evolving algorithms and workload needs



(Source: Xilinx)

Questions

How do FPGAs accelerate AI and ML through custom hardware design?

- A. Through pipelining the data
- B. Custom multiply-accumulate operations in parallel
- C. Increasing CPU clock speed to allow data to be processed faster
- D. Storing more data in RAM to enable faster parallel processing

How do programmable interconnects help an FPGA work?

- A. By permanently storing the program instructions
- B. They route signals between blocks so the FPGA can form various circuitry
- C. They control power delivery to the chip
- D. By getting rid of the need for logic blocks

T/F FPGAs are always on the same chip as the CPU?

What is the primary difference between developing on an FPGA vs a CPU?

- A. FPGAs are easier
- B. CPUs are easier
- C. FPGA development adapts the architecture while CPU development adapts the program
- D. CPU development adapts the architecture while FPGA development adapts the program

Links / Works Cited

- <https://medium.com/@mike.anderson007/the-role-of-fpgas-in-ai-acceleration-36029fc79faa>
- [Innovation in Action: Building FPGA-based Accelerators for Cutting-Edge Applications](#)
- [Fundamentals of FPGA Based Acceleration](#)
- <https://www.reflexces.com/newsroom/why-are-fpgas-faster-than-cpu>
- <https://www.vmaccel.com/posts/the-fpga-cloud-advantage>
- <https://www.intel.com/content/www/us/en/docs/oneapi-fpga-add-on/developer-guide/2024-1/fpga-architecture-overview.html>
- <https://www.xilinx.com/publications/events/developer-forum/2018-frankfurt/fundamentals-of-fpga-based-acceleration.pdf>
- <https://www.logic-fruit.com/blog/fpga/fpga-design-architecture-and-applications>
- <https://www.logic-fruit.com/blog/fpga/fpga-vs-cpu/>

Optical/Photonic Computing

By: Aaron, Andrew, Connor, and David

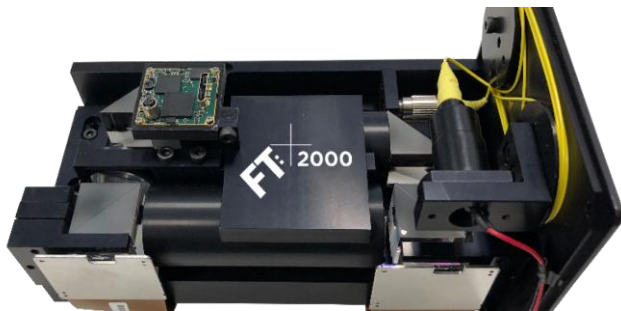
What is optical computing?

Originally developed in the 1980's, the concept of optical computing, sometimes referred to as photonic computing, uses light and the flow of photons for computing instead of electricity

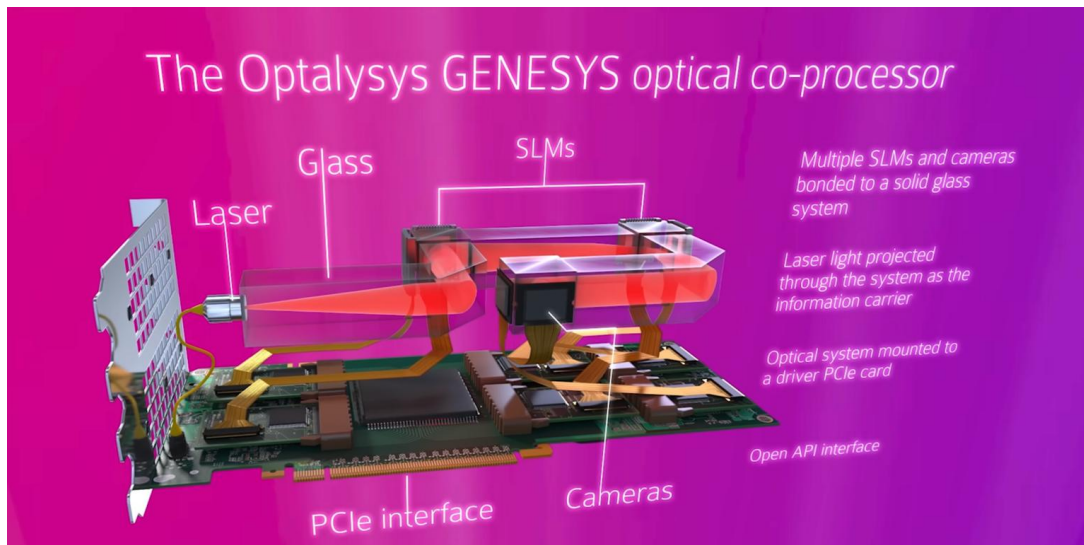
The light used in this computation is invisible (450nm - 700nm wavelength)

Wave propagation(the interference of wave patterns) is used to determine a result in a computation

Optical computing works by modulating properties of light, such as **intensity**, **phase**, **polarization**, and **wavelength**, to encode and process information



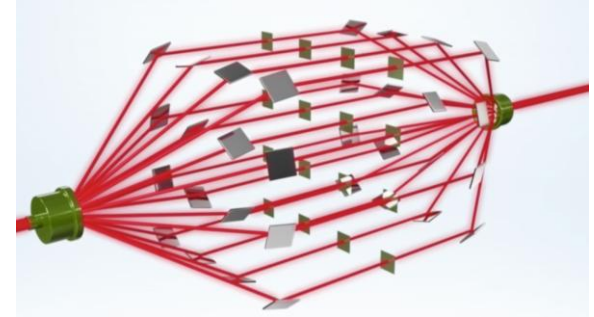
[Scientific Computing World](#)



[Futurology Youtube](#)

Advantages

- Generates negligible heat unlike electronic computers we have today
- Can transfer data with very low power consumption
- Scaling optical circuits is easier and cheaper than traditional computing methods
- Can operate with switch times as fast as a few picoseconds
- In particular cases, optical computing can essentially eliminate latency experience in electronic computing



[Futurology Youtube](#)



[Futurology Youtube](#)

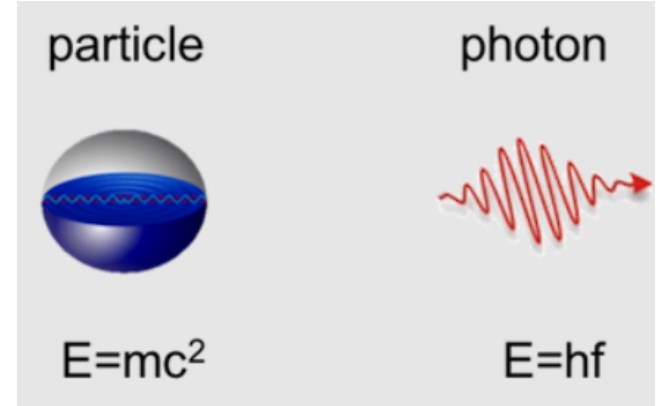
X1 Photonic Module vs Electronic Cable (1.2 Terabit)

Disadvantages

- Weak nonlinearity
 - Photons have no **mass** or **charge**
 - No photon-photon interaction
- Difficult to store memory
 - Photons never stop moving
- Hardware size constraints

[Photon Creation and Absorption - EWT](#)

- Optical switches are 100-1000 times larger than a transistor
- OEO conversion(Optical-Electrical Optical Conversion)
 - Requires lots of energy to convert
 - All photonic systems must perform OEO



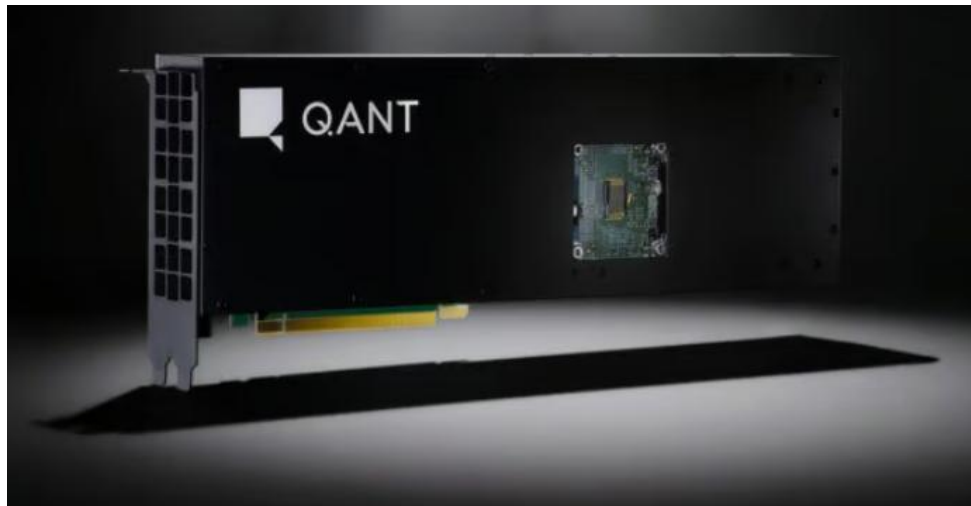
Applications

- **AI Data Centers**
 - Because it's faster and more energy efficient it would be an incredible help to data centers.
- **Quantum Computing**
 - Quantum computers can require temperatures colder than deep space, photonic quantum operations can happen at room temperature.
- **Complex Calculations**
 - The diffractions of light already can mimic complex equations. That allows them to be used to solve complex equations with less steps.

Current Products: Q.ANT's NPU Gen 2

NPU Gen 2:

- Speed-up to 8 GOPS (Giga Operations Per Second)
- Enhanced nonlinear processing capabilities
- Setting the conditions for multidimensional optical scaling
- Announced March 17 2026



[Q.ANT Unveils its Second-Generation Photonic Processor to Power the Next Wave of AI and HPC - Q.ANT](#)

More on Q.ANT's NPU (Native Processing Unit) Gen 2

- First commercial photonic processor
- Promises up to 30 times the energy efficiency of conventional CMOS technologies
- Reduces operational costs and the environmental impact of data centers
- Tackles the problem with conventional compute technologies reaching their limits
- Help reduce energy consumption as AI surges
- “Complex mathematical models for AI training and inference, machine learning, physics simulations and time-series analysis can be executed with exceptional performance.”

More on Q.ANT's NPU Gen 2

“Chip manufacturing is geographically and structurally concentrated.

The current landscape of AI hardware production is centralized in few regions and dominated by a small number of companies. Q.ANT's photonic chips can be produced on refurbished semicon fabrication lines enabling cost-efficient and decentralized chip production”

[Q.ANT Unveils its Second-Generation Photonic Processor to Power the Next Wave of AI and HPC - Q.ANT](#)



[Q.ANT Unveils its Second-Generation Photonic Processor to Power the Next Wave of AI and HPC - Q.ANT](#)

Exam Question 1

1. Optical computing uses what 4 properties of light to encode and process data?

a. Intensity, phase, polarization, and wavelength

b. Phase, reflection, wavelength, and refraction

c. Intensity, phase, polarization, and reflection

d. Intensity, polarization, phase, and diffraction

Exam Question 2

2. What is the size difference between optical switches and transistors?
- a. Optical switches are 1 - 10 times smaller
 - b. Optical switches are 50 times larger
 - c. Optical switches are 2,000,000 times smaller
 - d. Optical switches are 100 - 1000 times larger

Exam Question 3

3. Which of the following is NOT an advantage of optical computing?

negligible heat, fast switch times, easy to store, or easy to scale

a. Negligible heat

b. Fast switch times

c. Easy to store

d. Easy to scale

Exam Question 4

4. Q.ANT's NPU Gen 2 promises what compared to the efficiency of conventional CMOS technologies?

a. Increase its energy efficiency

b. Increase speed

c. Decrease size constraints

d. Reduce production price

Recommended Readings & Videos

Readings:

[Optical Computing Basics](#)

[Silicon Photonics](#)

[Photonics and Quantum Computing](#)

[Will Photonics Revolutionize Computing?](#)

[Trends and challenges in Photonics and Optical IO](#)

[Why Optical Computing isn't Mainstream](#)

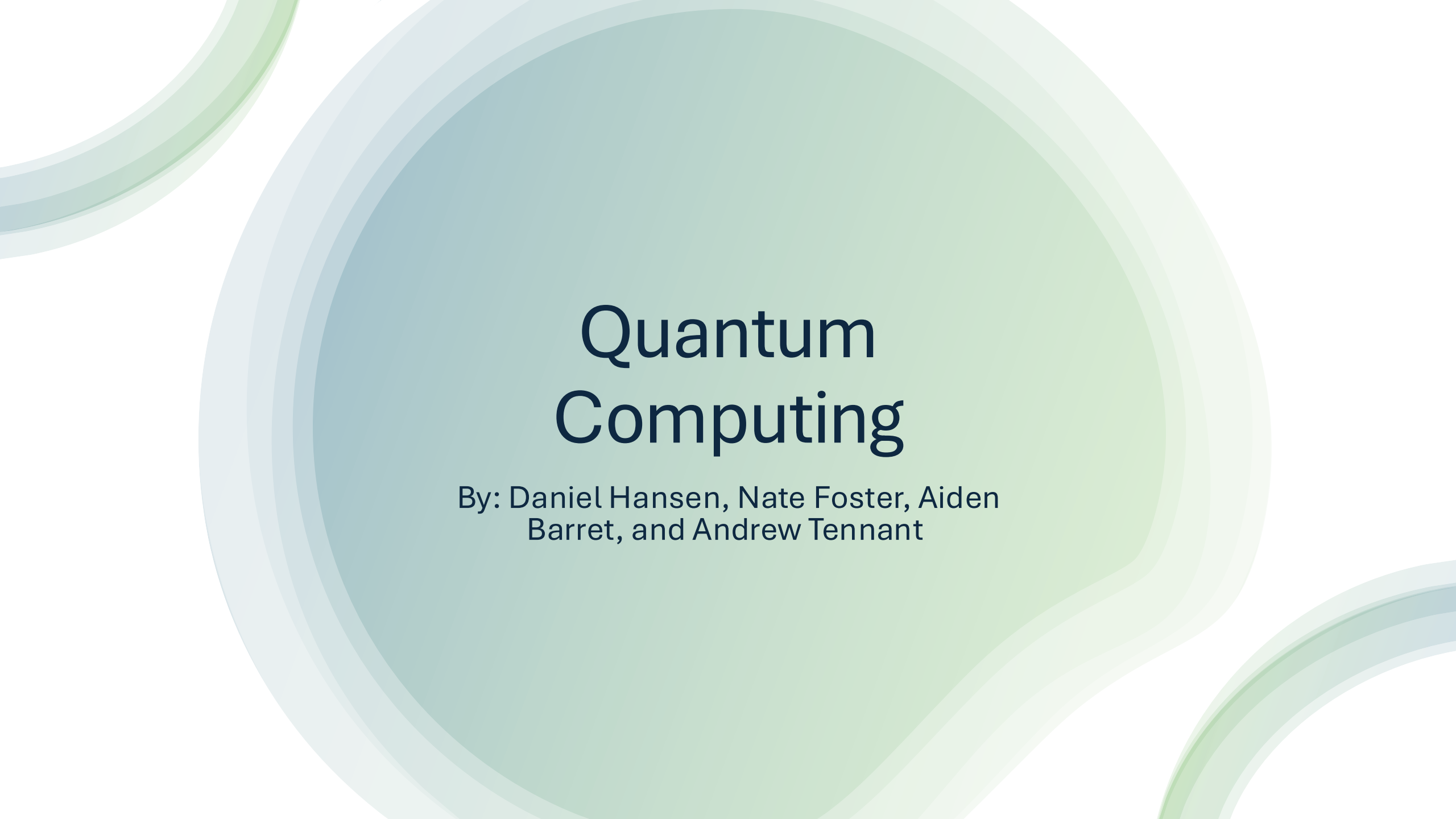
[Q.ANT Unveils its Second-Generation Photonic Processor to Power the Next Wave of AI and HPC - Q.ANT](#)

Videos:

[What is optical computing?](#)

[Akhetonics - Photonic Chips](#)

Questions?

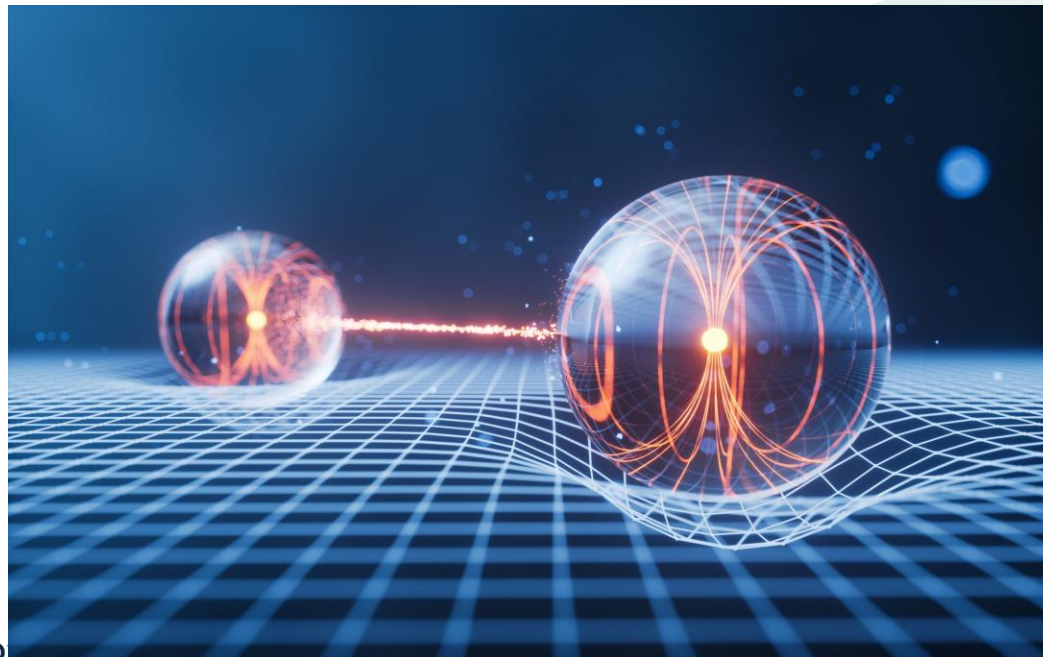


Quantum Computing

By: Daniel Hansen, Nate Foster, Aiden
Barret, and Andrew Tennant

Superposition & Entanglement

- Qubits versus Bits
 - Superposition
 - The ability of qubits to be both 0 and 1.
 - Entanglement
 - When two qubits are linked so that the state of one influences the state of the other.
- Decoherence
 - Quantum states are fragile. They must be kept in very cold environments or decoherence (signal noise) will ruin the calculation.

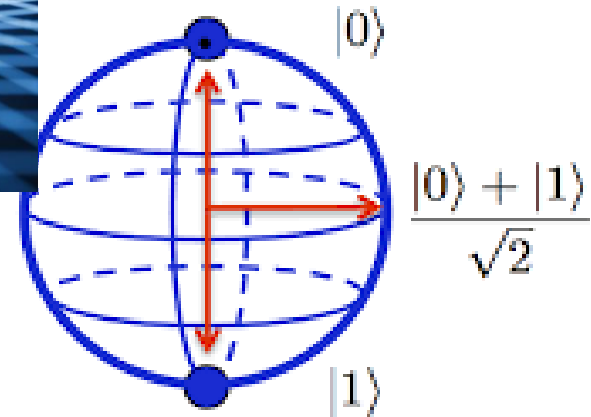


UC Santa Barbara



1

Classical Bit



Qubit

<https://poetryinphysics.wordpress.com>

Cryptography

- No-cloning theorem
 - If you try to read data in a quantum state, a wave function collapse will cause it to change.
- Quantum Key Distribution
 - Use quantum entanglement to generate a key
 - Uses quantum mechanics instead of computational difficulty
 - Relies on already having secure communication



security-insider.de

How do we build one?

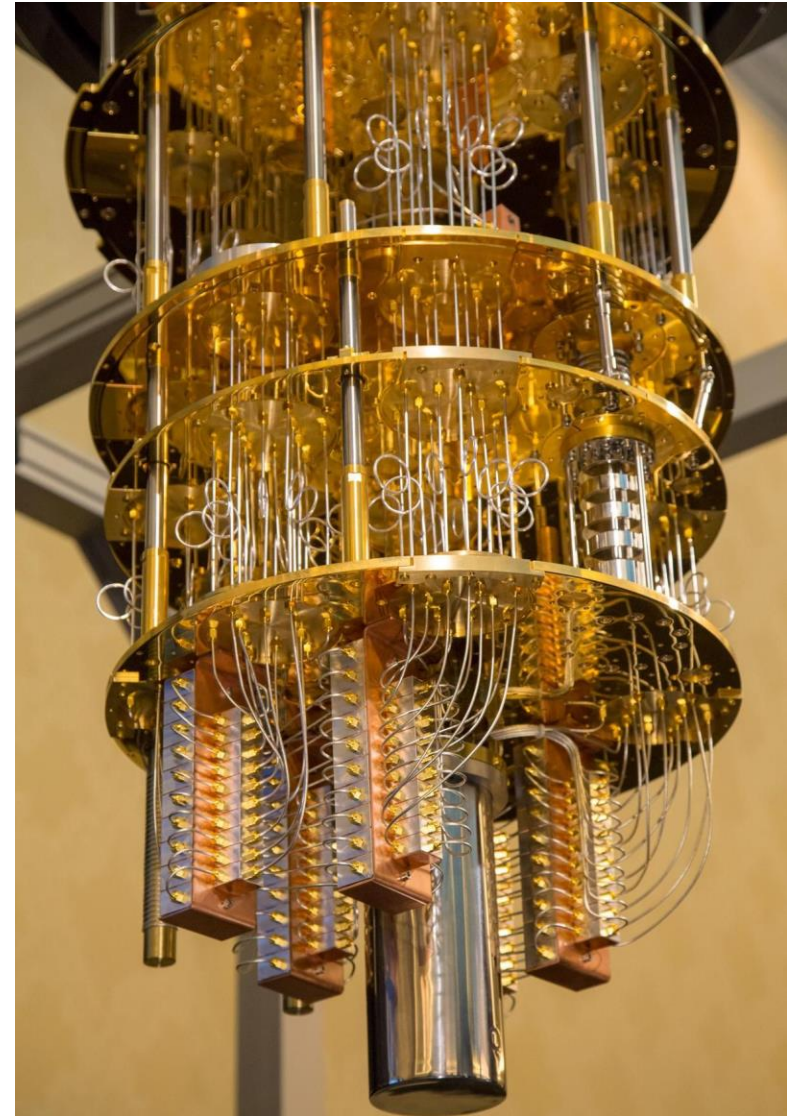
- Chandelier
 - Limits mechanical vibrations
 - Maintains thermal isolation
- Cryptography
 - Keeps Qubits stable
 - Superconductors
- Quantum Amplifiers
 - Signal Integrity
 - Error Correction



Screenstitchy - Screen Video Recorder is sharing your screen. Stop sharing Help

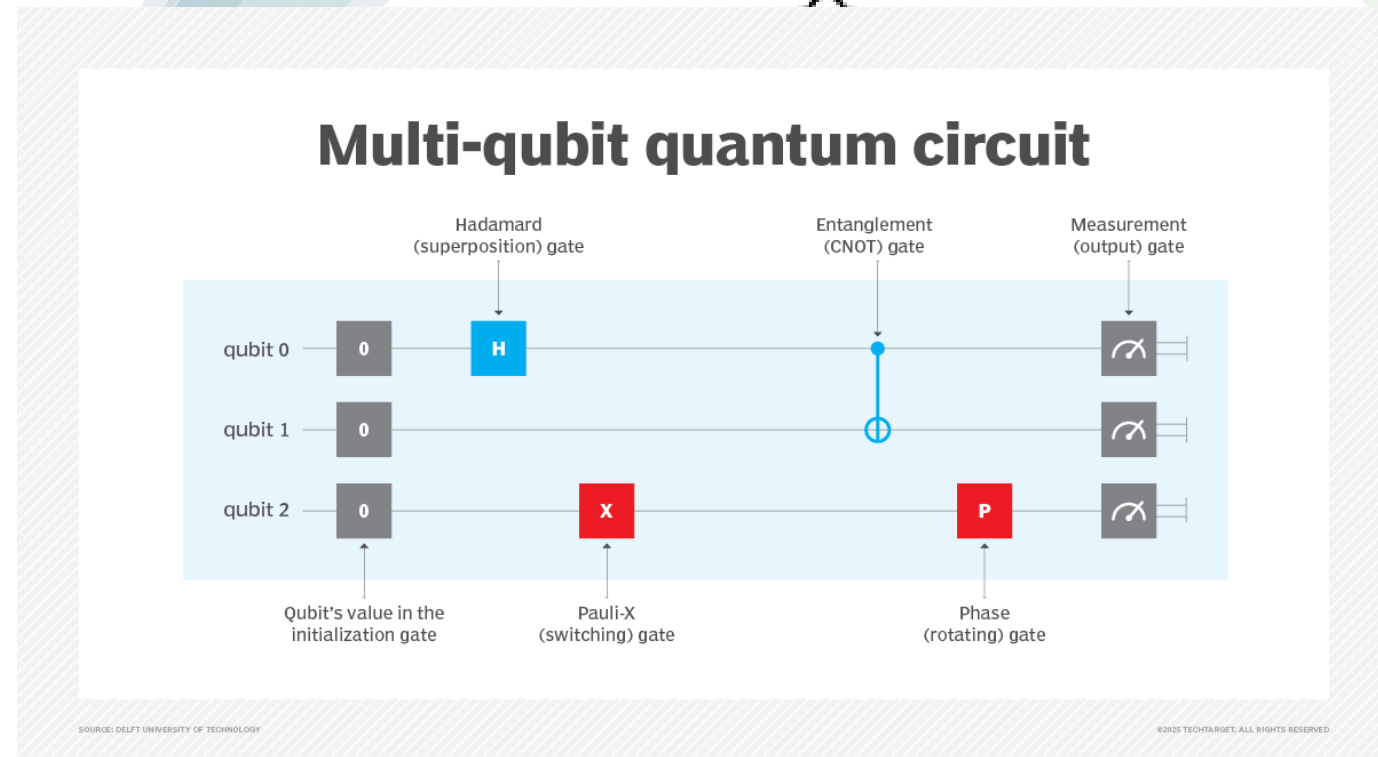
How do we build one?

- Chandelier
 - Limits mechanical vibrations
 - Maintains thermal isolation
- Cryogenics
 - Keeps Qubits stable
 - Superconductors
- Quantum Amplifiers
 - Signal Integrity
 - Error Correction

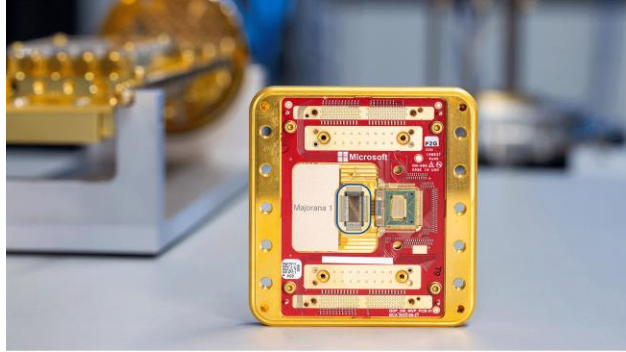


How do we build one?

- Gates need to be able to:
 - Be reversible
 - Preserve the total probability
 - Have qubits exist in multiple states
- Types of Quantum Gates:
 - Single Qubit
 - Pauli-X, Y, or Z
 - Hadamard
 - Multiple Qubit
 - Controlled Not (CNOT)



Quantum vs. Classical Computing



Majorana 1, Microsoft

Quantum

- Excels at complex problems with many variables.
 - Molecular Simulation
 - Cryptography
 - Financial and Logistics Modeling
 - Machine Learning
- Problems with many inputs and many possible outcomes



AMD

Classical

- Linear tasks
 - Web browsing
 - Standard applications
 - Spreadsheet processing
- Orchestration of many tasks
- Very low error rate
- Excels in the everyday use case
- Problems with few possible outcomes

Questions

In quantum computing what property allows a qubit to exist in a state of both 0 and 1 until the point that it is measured?

A. Entanglement B. Superposition C. Decoherence D. Interference

What is it called when two qubits affect each other's state no matter how far apart they are?

A. Quantum Tunneling B. Entanglement C. Connection D. Interference

True/False: Quantum computing is faster than classical computing in every use case.

What gate changes the phase angle of the qubit?

A. CNOT Gate B. Pauli Gate C. Hadamard Gate D. Adjoint

Recommended Readings

- <https://www.ibm.com/think/topics/quantum-computing>
- [Why Do Quantum Computers Look So Weird?](#)
- Quantum Computing: Navigating the Future of Computation, Challenges, and Technological Breakthroughs <https://www.mdpi.com/2624-960X/6/4/39>
- <https://www.spinquanta.com/news-detail/decoherence-in-quantum-computing-everything-you-need-to-know>
- <https://www.spinquanta.com/news-detail/quantum-computing-explained-in-simple-terms-a-complete-beginners-guide-2026>
- <https://www.csiro.au/en/news/All/Articles/2025/January/2025-huge-advances-in-quantum-computing>
- <https://phys.org/news/2025-09-scientists-quantum-unconditionally-outperform-classical.html>